

REVIEW ARTICLE

Neuropsychological differential diagnosis of Alzheimer's disease and Lewy body dementia: A systematic review

T. Julio-Ramos^a, V. Mora-Castelleto^b, C. Fonca-González^b, C. Adames-Valencia^c,
I. Cigarroa^d, C. Méndez-Orellana^e, D. Toloza-Ramírez^{f,g,*}

^a PhD Program in Health Sciences and Engineering, Universidad de Valparaíso, Valparaíso, Chile

^b Fonoaudióloga, Pontificia Universidad Católica de Chile, Santiago, Chile

^c Fonoaudióloga, Centro Vida Sur, Puerto Montt, Chile

^d Escuela de Kinesiología, Facultad de Ciencias de la Salud, Universidad Católica Silva Henríquez, Santiago, Chile

^e Carrera de Fonoaudiología, Departamento Ciencias de la Salud, Facultad de Medicina, Pontificia Universidad Católica de Chile, Santiago, Chile

^f Exercise and Rehabilitation Sciences Institute, School of Speech Therapy, Faculty of Rehabilitation Sciences, Universidad Andres Bello, Santiago, 7591538, Chile

^g Interdisciplinary Center for Neuroscience, Faculty of Medicine, Pontificia Universidad Católica de Chile, Santiago, Chile

Received 28 November 2022; accepted 29 March 2023

KEYWORDS

Alzheimer's disease;
Dementia with Lewy
bodies;
Differential
diagnosis;
Neuropsychological
assessment;
Neural correlate

Abstract

Introduction: Alzheimer's disease (AD) reports heterogeneity of neuropsychological symptoms misleading the differential diagnosis with other forms of dementia, such as dementia with Lewy bodies (DLB). About 50% of DLB patients are misdiagnosed as AD cases. Likewise, the diagnosis of both diseases is mainly based on clinical characteristics. However, differentiating AD of those with DLB based on neuropsychological symptoms and anatomical and functional brain changes remains challenging.

Aim: To establish the main neuropsychological, anatomical, and functional similarities and differences in patients with AD and DLB.

Methods: The present study followed the PRISMA guidelines and included studies from the PubMed, Scopus, and Web of Sciences databases, published between January 2000 and July 2022.

Results: 41 articles were included in this systematic review for critical analysis. Our results suggest that the cognitive key domains to consider in the differential diagnosis are memory, executive function, attention, visuospatial/visuoconstructive skills, and verbal fluency (both semantic and phonological). The stage and severity of both diseases would be essential for

DOI of refers to article: <https://doi.org/10.1016/j.nrl.2023.03.006>.

* Corresponding author.

E-mail address: david.toloza@me.com (D. Toloza-Ramírez).

<https://doi.org/10.1016/j.nrleng.2025.10.001>

2173-5808/© 2024 Sociedad Española de Neurología. Published by Elsevier España, S.L.U. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

PALABRAS CLAVES

Enfermedad de Alzheimer;
Demencia con Cuerpos de Lewy;
Diagnóstico diferencial;
Neuropsicología;
Redes neurales

differential diagnosis. On the other hand, the anatomical and functional changes suggest a similar atrophy pattern between AD and DLB in the frontal, parietal, temporal, hippocampal, and precuneus regions.

Conclusion: The differential diagnosis between AD and DLB is challenging in clinical practice. Therefore, our results suggest exploring cognitive linguistic markers along with correlating these markers with anatomical and functional brain changes.

© 2024 Sociedad Española de Neurología. Published by Elsevier España, S.L.U. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Diagnóstico neuropsicológico diferencial en enfermedad de Alzheimer y demencia por cuerpos de Lewy: una revisión sistemática

Resumen

Introducción: La Enfermedad de Alzheimer (EA) reporta heterogeneidad de síntomas neuropsicológicos. Esto conduce a errores diagnósticos con otras formas de demencia como la demencia por cuerpos de Lewy (DPCL). De hecho, alrededor del 50% de los pacientes con DPCL son confundidos como casos de EA. Si bien el diagnóstico de ambos cuadros se basa principalmente en aspectos clínicos, continúa siendo un desafío su diferenciación en base a los síntomas neuropsicológicos y los patrones de atrofia cortical.

Objetivo: Establecer las principales similitudes y diferencias neuropsicológicas y de atrofia cortical en pacientes con EA y DPCL.

Metodología: La presente revisión sistemática siguió los lineamientos establecidos en la declaración PRISMA, utilizando las bases de datos PubMed, Scopus y Web of Science. La búsqueda estuvo limitada a estudios observacionales analíticos de pruebas diagnósticas, publicados en idioma inglés entre Enero 2000 y Julio 2022.

Resultados: La búsqueda dio como resultado 41 artículos finales. Del total de los artículos identificados se sugiere que los elementos neuropsicológicos claves para el diagnóstico diferencial entre EA y DPCL son la memoria, función ejecutiva, atención, habilidades visuoespaciales/visuoconstructivas, y fluidez verbal (semántica y fonológica) siendo el estadio y grado de severidad de cada cuadro críticos en el proceso diagnóstico. Los resultados además sugieren un patrón de cambios anatomo-funcionales similar entre EA y DPCL en áreas frontal, parietal, temporal, hipocampal y precuneus.

Conclusión: El diagnóstico diferencial entre EA y DPCL es un desafío en la práctica clínica por lo que esta revisión propone explorar marcadores cognitivos con énfasis en los indicadores lingüísticos además de los cambios anatómicos y funcionales de las áreas cerebrales.

© 2024 Sociedad Española de Neurología. Publicado por Elsevier España, S.L.U. Este es un artículo Open Access bajo la CC BY-NC-ND licencia (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Alzheimer disease (AD) is the leading neurodegenerative cause of dementia, accounting for 60%–70% of all cases.^{1–3} Advances in neuroimaging techniques have enabled the detection of molecular alterations in the brains of patients with AD, with high levels of sensitivity and specificity; specifically, positron emission tomography (PET) studies⁴ can detect extracellular β -amyloid ($A\beta$) plaques and intracellular neurofibrillary tangles of hyperphosphorylated tau protein⁵ in frontal, temporal, and parietal regions of the brain.^{6,7} The most common methods for early detection of AD include $A\beta$ -PET, tau-PET, [^{18}F]-fluorodeoxyglucose PET (FDG-PET), and volumetric analysis of brain MRI images. In addition to these neuroimaging techniques, cerebrospinal fluid (CSF) biomarkers of $A\beta$ ($A\beta_{42}/A\beta_{40}$ ratio,

$A\beta_{42}$) and phosphorylated tau (p-tau) have been implemented in clinical practice to support the diagnosis of AD.⁸

Although AD has been the main subject of interest in this area of research, such other diseases as frontotemporal dementia (FTD) and Lewy body dementia (LBD) are also relevant.^{9–11} Toloza-Ramírez et al.¹² highlight similarities between the symptoms of AD and behavioural variant FTD in terms of memory, executive function, and linguistic indicators.

LBD, in turn, is characterised by attention fluctuations,¹³ and accounts for 0.3%–24.4% of all cases of dementia.¹⁴ Neuropathological differential diagnosis considers α -synuclein deposition,^{13,15,16} as well as FDG-PET findings of hypometabolism in the occipital lobe, sparing the posterior cingulate.⁵

In neuropsychological terms, LBD manifests with severe impairment of memory, attention, executive function, and visuoconstructive skills, unlike AD, in which episodic memory is the main domain affected.^{17–19} Furthermore, both diseases present with language symptoms (e.g., verbal fluency), negatively affecting expressive language.^{20–23}

The heterogeneity of symptoms and patterns of atrophy in AD is a current subject of interest in cognitive neuroscience. Some studies suggest that LBD is often misdiagnosed and mistaken for AD.^{24–26} Likewise, post mortem studies reveal that approximately half of patients with LBD present a considerable burden of AD pathology; as a result, LBD is considered to be secondary to AD.^{13,27} Oda et al.²⁸ suggest that cognitive and emotional decline in LBD is explained by the loss of cholinergic neurons, as is also the case in AD.

Various lines of research have focused on the differential diagnosis between AD and LBD, studying the role of patterns of atrophy as a differentiating factor and seeking to identify neuropsychological deficits.^{29,30} Neuroimaging studies have demonstrated temporoparietal and occipital hypometabolism in both diseases; however, this sign tends to be milder in AD.³¹ Furthermore, the right cingulate cortex is reported to display greater connectivity in LBD, whereas in AD, the left hippocampus shows greater connectivity.³²

Some authors also describe neuropsychiatric symptoms that are key to differential diagnosis. Several studies suggest that hallucinations, agitation, and sleep alterations are more frequent in LBD than in AD; this is correlated with higher Neuropsychiatric Inventory (NPI) scores in patients with LBD.^{33–36} Furthermore, motor symptoms are observed in up to 85% of cases with LBD¹³; although these are not characteristic of typical AD, they do present in atypical cases of AD, such as corticobasal syndrome.³⁷ In fact, the evidence suggests that these neuropsychiatric symptoms promote greater functional and cognitive impairment in these patients, and even favour axonal degeneration and the pattern of atrophy in brain regions relevant to cognitive function (e.g., frontal and limbic areas).^{38–41}

Despite efforts to establish a consensus on the clinical and neuropathological characteristics involved in the differential diagnosis between AD and LBD,^{5,13} the neuropsychological deficits that support differential diagnosis remain unclear, particularly those related to cognition and language. Therefore, the aim of this review is to establish the main similarities and differences between AD and LBD in terms of neuropsychological symptoms (particularly in the language domain) and anatomical/functional changes. Thus, improving the assessment of cognitive domains will favour the design of early, appropriate interventions in patients with dementia, and particularly interventions targeting functional status in the activities of daily living.

Methods

This systematic review observed the criteria and flow diagram established in the PRISMA statement.⁴² The pro-

tocol was registered on the PROSPERO database (code CRD42021261164). The PRISMA checklist is included in the Supplementary material (Appendix 1).

Search strategy for identifying studies

The literature search was conducted on the following electronic databases, in the following order: PubMed, Scopus, and Web of Science. The search was limited to articles published between January 2000 and July 2022. This time period was selected due to the fact that research in the last 10 years has identified atypical profiles of AD; therefore, the search start date had to be extended to the year 2000 to obtain a general perspective of both typical and atypical AD. The general search strategy was as follows: (Alzheimer disease AND Lewy body dementia) AND (cognitive function OR memory OR executive function OR attention OR visuoconstructive skills OR visuospatial skills OR processing speed OR language) AND (neuropsychological assessment OR cognitive tests OR neuroimaging OR differential diagnosis). All terms were adapted to each database. The complete search strategy for each database is included in the Supplementary material (Appendix 2).

Study selection and inclusion/exclusion criteria

We applied the following inclusion criteria for screening by title and abstract: *a)* analytical observational studies of diagnostic tests, reporting neuropsychological assessment and/or neuroimaging study results; *b)* studies written only in English; and *c)* including patients aged at least 60 years with a clinical diagnosis of AD or LBD. We excluded the following types of article: *a)* editorials, experimental studies, systematic reviews with or without meta-analyses, study protocols, and theses; *b)* studies including patients with mixed dementia or pseudodementia; and *c)* studies including patients with history of psychiatric disorders (e.g., psychosis, schizophrenia).

Extraction of data

Studies were imported to the Rayyan software⁴³ to eliminate duplicate articles. Subsequently, 2 reviewers (VMC and DTR) applied the inclusion and exclusion criteria to the titles and abstracts of all articles. In the event that decisions could not be made based solely on the title and abstract, full texts of the articles were retrieved. A third reviewer (TJR) participated in the final selection of articles included for review, resolving disagreements through discussion where necessary.

Assessment of risk of bias and methodological quality

To evaluate the methodological quality and risk of bias of the articles selected, we used the QUADAS-2 tool,⁴⁴ an instrument designed and validated for the independent evaluation of methodological quality and risk of bias in studies of diagnostic accuracy. This instrument evaluates 4 key domains:

patient selection, the index test, the reference standard, and the patient flow and timing of the study (the latter 2 aspects are grouped together into the domain “flow and timing”). Studies were classified into one of 3 risk categories: high, low, or unclear. Methodological quality was assessed according to the domains patient selection, index test, and reference standard, and was classified as high, low, or unclear.

Data synthesis strategy

Table 1 presents a narrative synthesis of our findings. This synthesis summarises the general characteristics of the studies reviewed, such as study population, mean age, and the instruments/measures used. It also presents the main findings regarding the neuropsychological similarities and differences between AD and LBD, with the neuroimaging profiles reported.

Results

Literature search

The PRISMA flow diagram in Fig. 1 illustrates the article selection process.⁴² We identified the titles and abstracts of 447 articles, 50 of which were duplicates. We excluded 202 articles after application of the selection criteria. Of the 195 full-text articles screened for eligibility, 154 were excluded. Therefore, a total of 41 articles were included in the systematic review for qualitative analysis.

Imaging study characteristics

Table 1 shows the general characteristics of the 41 articles included for review. The year in which the greatest number of articles were published was 2017 (6 articles), with the years 2019, 2007, 2006, 2005, 2004, 2001 presenting the lowest number of articles (one each). The selected studies included a total of 8780 patients, with a mean age of 75.2 years (range, 60–80; standard deviation [SD]: 7.0). A total of 4498 (57%) were men.

Regarding the main neuropsychological findings, the results suggest that the cognitive domains of memory, attention, executive function, visuospatial/visuoconstructive skills, and verbal fluency (both semantic and phonological) are key aspects in the differential diagnosis of AD and LBD. From an anatomical/functional perspective, the pattern of atrophy in the frontal, parietal, and temporal lobes, as well as the hippocampus and precuneus, seems to enable differential diagnosis between the 2 disorders. However, the results underscore that the level of functional connectivity is reduced in sensorimotor, temporal, basal ganglia, thalamic, insular, and anterior cingulate networks in LBD, whereas in AD it is reduced in frontal and temporal areas, with increased connectivity in the posterior cingulate and hippocampus. This is a relevant finding, as this

may be a useful indicator for establishing an early, accurate diagnosis.

Finally, the neuropsychological instruments and measures most frequently used to diagnose AD and LBD were the Mini-Mental State Examination (MMSE), Montreal Cognitive Assessment (MoCA), Boston Naming Test (BNT), Trail-Making Test (TMT) parts A and B, Frontal Assessment Battery (FAB), FAS verbal fluency test, and Cambridge Cognition Examination (CAMCOG). The most frequently used neuroimaging techniques were MRI, fMRI, and PET, which are combined to establish an anatomical-functional correlate (Table 1).

Risk of bias and methodological quality assessment

The 41 articles selected were analysed using the QUADAS-2 tool (Fig. 2).⁴⁴ In the risk of bias assessment, the domains patient selection and flow and timing presented low risk of bias in 95% of studies and unclear risk in 5%. In the index test domain, 90% of studies presented low risk, 5% presented high risk, and 5% presented unclear risk. In the reference standard domain, 93% of studies presented low risk, 5% presented high risk, and 2% presented unclear risk. In turn, in the assessment of methodological quality, 95% of articles presented high quality for patient selection, compared to 90% and 93% for index test and reference standard, respectively.

Discussion

The objective of this review was to establish the main neuropsychological and anatomical/functional similarities and differences between AD and LBD. Our results suggest that memory, attention, executive function, visuospatial/visuoconstructive skills, and semantic and phonological verbal fluency may be considered additional indicators supporting differential diagnosis. Several studies^{84–87} report differences in the progression of overall cognitive impairment in AD and LBD, suggesting that progression is faster in LBD. In fact, Gilil and Aarsland⁸⁸ report that LBD progresses to advanced cognitive impairment within 5 years of follow-up. They also note that the main characteristic of AD is the heterogeneity of the cognitive domains affected. Regarding anatomical/functional changes, specific patterns of atrophy enabling differentiation between AD and LBD are reported in frontal, parietal, and temporal regions, the hippocampus, and the precuneus.

Episodic memory

Our findings suggest that episodic memory is impaired in both conditions. However, the deficit is more severe in AD; this is consistent with the results of previous studies.^{13,89} In contrast, Calderon et al.²⁰ report that episodic memory impairment does not differ between the 2 diseases; in fact, they suggest that working memory is more severely

Table 1 Summary of the characteristics of the reviewed studies.

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Buciuc et al. ⁴⁵ (2021)	Total sample of 20 participants: 6 with logopenic variant primary progressive aphasia and LBD, 7 with logopenic variant primary progressive aphasia and typical AD, 5 with logopenic variant primary progressive aphasia and hippocampal variant AD, and 5 with logopenic variant primary progressive aphasia and frontotemporal lobar degeneration	61.3 (—)	✓	✓	✓	4 (20)/16 (80)	MMSE MoCA MDS-UPDRS III FBI NPI WAB BNT PPT TT WMS-III VOSP TMT-A D-KEFS ST ROCF FDG-PET PiB-PET MMSE CDR DS TMT-A TMT-B eTIV MRI	Patients with logopenic variant primary progressive aphasia and LBD displayed global cognitive impairment affecting memory, attention, executive function, and semantic and phonological verbal fluency; similar deficits are reported in patients with typical AD and the same type of aphasia. However, severity was greater in the first group. From a linguistic perspective, patients with LBD presented severe anomia and deficits in phonological fluency; this would be a differentiating factor with respect to the typical AD group. From an anatomical/functional perspective, patients with LBD presented moderate-severe neuronal loss in the substantia nigra; this sign was milder in the typical AD group. Greater parietal hypometabolism was observed in patients with LBD, whereas in AD the pattern of atrophy is heterogeneous, affecting temporal, parietal, and occipital areas. In neuropsychological testing, patients with LBD displayed severe impairment of working and short-term memory, as well as significant alterations to executive function and attention performance. Marked atrophy was observed in the pulvinar nucleus, particularly in the left medial region; this alteration occurs at early stages, promoting executive deficits in LBD.
Tak et al. ⁴⁶ (2020)	Total sample of 76 participants: 38 with LBD and 38 controls	74.9 (6.2)	✓	✓	—	38 (50)/38 (50)		

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Schumacher et al. ⁴⁷ (2018)	Total sample of 91 participants: 31 with LBD, 29 with AD, and 31 controls	76.6 (7.5)	✓	✓	—	30 (33)/61 (67)	MMSE CAMCOG UPDRS-III CAF	Neuropsychological evaluation suggested slight cognitive differences between LBD and AD, with the latter showing greater impairment of orientation to space and time, expressive language (verbal fluency), recent and distant memory, learning, attention, executive function, and perceptual abilities. Patients with LBD presented impaired functional connectivity of sensorimotor, temporal, basal ganglia, thalamic, insular, and anterior cingulate networks. AD was associated with reduced connectivity in frontal and temporal areas. However, comparison between the 2 revealed slight differences in frontal and temporal atrophy, which are key to the initial diagnosis of both diseases.
Donaghy et al. ⁴⁸ (2018)	Total sample of 77 participants: 37 with LBD, 20 with AD, and 20 controls	75.9 (6.9)	✓	✓	—	14 (18)/63 (82)	NPI fMRI MRI PET CT CSF ACE-R RAVLT TMT-A TMT-B FAS GNT IADL BADL NPI DCFS CAF GDS MDS-UPDRS motor score BP	Patients with LBD and positivity for beta amyloid present impaired orientation to time and place, attention, memory, semantic and phonological verbal fluency, visuospatial skills, delayed recall, and executive function. However, these deficits tend to be less pronounced in patients who are not positive for beta amyloid, suggesting that they may play a key role in the differential diagnosis between LBD and AD. AD was associated with greater atrophy of frontal, temporal, and parietal regions. Compared with AD, LBD presented greater volume of the hippocampus and medial temporal lobe, as well as greater orbitofrontal perfusion, which was not observed in AD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Elder et al. ⁴⁹ (2017)	Total sample of 204 participants: 65 with LBD, 76 with AD, and 63 controls	77.8 (6.8)	✓	✓	—	75 (37)/129 (63)	MMSE CAMCOG UPDRS-III NPI CAF MRI CSF	No significant differences in overall cognitive function were observed between AD and LBD; however, patients with AD presented better memory performance. Patients with LBD presented moderate atrophy of temporal, hippocampal, parahippocampal, and entorhinal cortical regions, compared to severe atrophy in AD.
Kemp et al. ⁵⁰ (2017)	Total sample of 66 participants: 37 with LBD and 29 controls	68 (8.3)	✓	✓	—	33 (50)/33 (50)	MRI IADL ADL MMSE FCSRT DMS-48 DS FAB TMT-A TMT-B DSB DMS-48 DO80 FLE DSym SG FSE ROCF VOSP Mini-SEA FPRT FER RMET	Patients with LBD displayed overall impairment of cognitive function. The memory domains of free recall and delayed recall were particularly affected. Furthermore, executive function, language (naming and semantic fluency), and visuoconstructive skills displayed severe impairment in this group. From an anatomical/functional perspective, patients with LBD presented bilateral hippocampal atrophy and positivity for beta amyloid and tau markers.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Peraza et al. ⁵¹ (2016)	Total sample of 53 participants: 19 with LBD, 18 with AD, and 16 controls	76.2 (6.9)	✓	✓	✓	12 (23)/41 (77)	MRI fMRI MMSE UPDRS CAMCOG	Patients with LBD and AD displayed impairment in the cognitive domains of remote memory, recent memory, episodic and short-term memory, orientation, language (semantic and phonological verbal fluency), attention, abstract reasoning, executive function, and visuoconstructive skills. Although no significant differences were observed, neuropsychological test scores were lower in patients with AD. The patterns of brain involvement in LBD and AD were similar in such areas as the precuneus, posterior cingulate, and frontal lobe. Furthermore, patients with LBD displayed greater atrophy in the cerebellum, right precuneus, left lingula, left thalamus, and left postcentral gyrus. Compared to LBD, AD was associated with greater impairment in the domains of memory, language, attention, perception, and abstract reasoning. However, executive function was severely impaired in LBD. These patients also showed greater impairment of phonological verbal fluency compared to those with AD. The anatomical/functional study identified increased functional connectivity in the posterior cingulate in AD and decreased connectivity in the left medial prefrontal cortex in LBD.
Kobeleva et al. ⁵² (2017)	Total sample of 71 participants: 30 with LBD, 20 with AD, and 21 controls	75.4 (6.8)	✓	✓	—	13 (18)/58 (82)	CAF NPI MMSE CAMCOG FAS UPDRS CDS CAF NPI fMRI	

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Schumacher et al. ⁵³ (2019)	Total sample of 89 participants: 39 with LBD, 28 with AD, and 22 controls	76 (6.3)	✓	✓	—	18 (20)/71 (80)	MMSE CAMCOG UPDRS CAF NPI ANT VBM MRI	Patients with LBD present less marked cognitive impairment than those with AD. Cognitive impairment is more evident in AD, affecting the domains of orientation, recent memory, remote memory, learning, language comprehension and speech, verbal fluency, attention, perception, and abstract reasoning. Similarly, patients with LBD presented more frequent cognitive fluctuations than those with AD. Furthermore, the pattern of atrophy in AD showed lower grey matter volume in the temporal, lingual, and left frontal cortex, as well as the precuneus. In contrast, LBD was associated with greater atrophy in the frontal cortex, particularly in the right hemisphere.
Heitz et al. ⁵⁴ (2016)	Total sample of 64 participants: 33 with LBD, 15 with AD, and 16 controls	69.1 (10.0)	✓	✓	—	33 (52)/31 (48)	MMSE IADL CAF FCSRT DMS-48 DO80 FAB DS TMT-A TMT-B DSST FLE ROCF VOSP MRI VBM Mini-SEA RMET PF SF	Patients with AD presented impairment of immediate and delayed recall, attention, executive function, and semantic verbal fluency. In contrast, patients with LBD showed deficits in phonological verbal fluency, praxis, and abstract reasoning. Patients with AD presented greater atrophy in the left medial frontal gyrus, bilateral parietal lobe (including the precuneus), right cuneus, left lingual gyrus, and bilateral medial temporal lobe. Patients with LBD presented greater atrophy of the left cingulate gyrus and right medial frontal gyrus.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Chabran et al. ⁵⁵ (2020)	Total sample of 159 participants: 79 with LBD, 58 with AD, and 22 controls	70.2 (8.5)	✓	✓	—	83 (52)/76 (48)	MMSE MCFS MRI fMRI VBM DMN FPN DAN MoCA	Patients with LBD and AD presented significant impairment of general cognition; however, AD was associated with greater involvement of memory, attention, visuoconstructive skills, and executive function. From an anatomical/functional perspective, patients with AD presented greater atrophy of the medial temporal lobe, including the parahippocampal gyrus, hippocampus, and amygdala. In LBD, greater atrophy was also observed in temporal lobe structures, as well as the insula and frontal lobes.
Yamamoto et al. ⁵⁶ (2017)	Total sample of 130 participants: 57 with AD and 73 with LBD	72.6 (8.7)	✓	✓	—	99 (76)/31 (24)		Patients with AD presented greater impairment of visuoconstructive skills and semantic and phonological verbal fluency. LBD was associated with significant memory impairment, particularly affecting delayed recall. Compared to AD, LBD was associated with a greater deficit in naming.
Azar et al. ³⁰ (2020)	Total sample of 51 participants: 34 with AD and 17 with AD + LBD	74.7 (8.4)	✓	✓	✓	29 (59)/22 (41)	MMSE CDR Modified MMSE HVLT BNT CFL VF TMT-A TMT-B PC CSF CT MRI	Patients with AD performed better for visuoconstructive skills and information processing speed. Regarding language, the AD group performed better for phonological verbal fluency. Therefore, these 3 domains are proposed as factors for study in the differential diagnosis of AD and LBD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Firbank et al. ⁵⁷ (2016)	Total sample of 78 participants: 23 with LBD, 32 with AD, and 23 controls	75.7 (6.7)	✓	✓	—	16 (20)/62 (80)	MMSE UPDRS CAMCOG MCFS CAF NPI FAS ANT DMN fMRI MMSE	No significant differences in overall cognitive impairment were observed between AD and LBD. However, LBD was associated with greater impairment of executive function and poorer performance in phonological verbal fluency tasks. The anatomical/functional study revealed similar involvement in both groups in the frontal, parietal, and occipital lobes, with an impact on the connectivity of these areas.
Peraza et al. ⁵⁸ (2015)	Total sample of 54 participants: 18 with LBD, 19 with AD, and 17 controls	76.2 (6.8)	✓	✓	—	11 (20)/43 (80)	UPDRS CAMCOG CAF NPI fMRI	Compared to LBD, AD was associated with greater deficits in the cognitive domains of orientation, recent memory, remote memory, learning, concentration, perception, abstraction, and language (comprehension and expression). From an anatomical/functional perspective, patients with LBD presented greater involvement of parietal and posterior occipital regions. In contrast, the most affected areas in AD were the temporal cortex, right occipital lobe, and right frontal lobe.
Brenowitz et al. ⁵⁹ (2017)	Total sample of 1603 participants: 193 without major neuropathology, 195 with stroke, 110 with LBD, 51 with LBD + stroke, 450 with AD, 292 with AD + LBD, 217 with AD + stroke, and 95 with AD + LBD + stroke	83 (9.9)	✓	✓	✓	706 (44)/897 (56)	CDR LMS-A DS SF PF BNT DSym TMT-A TMT-B MRI	Patients with AD present greater impairment of episodic and working memory than those with LBD. No significant differences between groups were observed for attention, executive function, or language (semantic verbal fluency and naming); however, faster decline was observed in patients with AD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Pizzi et al. ⁶⁰ (2015)	Total sample of 45 participants: 16 with AD, 16 with LBD, and 13 controls	75.6 (4.8)	✓	✓	—	23 (51)/22 (49)	CDR MMSE DRS FAB CAF UPDRS-III	In terms of overall cognitive performance, patients with AD presented greater deficits in the domains of memory, orientation to space and time, delayed recall, visuoconstructive skills, and executive function, compared to the LBD group. AD was associated with greater atrophy of the bilateral thalamus and temporal cortex. In contrast, the pattern of atrophy in LBD involved the bilateral thalamus and projections to the prefrontal, parietal, and occipital cortex bilaterally.
Jiménez-Huete et al. ⁶¹ (2014)	Total sample of 301 participants: 199 with AD, 65 with frontotemporal lobar degeneration, and 37 with LBD	75 (8.0)	✓	✓	✓	160 (53)/141 (47)	NPI MRI SPECT EEG MEC CFL PF TMT-A TMT-B CDT SS-IQCODE FAQ GDS-15 SRT	Patients with AD presented difficulties with naming, free recall, cued recall, and attention. Patients with LBD presented alterations in phonological verbal fluency, executive function, and visuoconstructive skills. Both diseases presented similar levels of impairment in semantic verbal fluency and short-term and working memory.
Nervi et al. ⁶² (2008)	Total sample of 591 participants: 157 with AD, 70 with LBD, and 364 controls	80 (7.6)	✓	✓	✓	396 (67)/195 (33)	CDR SRT BVRT RDT MMSE SF PF BNT BDAE WAIS-R UPDRS APOE-ε4 genotype	Patients with AD presented impairment of delayed recall, visuoconstructive skills, orientation, naming, and semantic and phonological verbal fluency, compared to patients with LBD. From a diagnostic perspective, no significant differences were observed between groups for presence of APOE-ε4.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings	
			Mean (SD)	(years)					
				60-70	71-80				> 80
Bradshaw et al. ⁶³ (2004)	Total sample of 25 participants: 13 with LBD and 12 with AD	78.3 (4.5)	✓	✓	✓	12 (48)/13 (52)	MMSE CDR NART UPDRS HADS BPRS FSIQ VIQ PIQ CAF WASI BI	Significant differences in neuropsychological test performance were observed between groups. Patients with AD presented greater memory impairment, whereas those with LBD presented greater impairment of attention. These deficits play a key role in patients' functional difficulties.	
Colloby et al. ⁶⁴ (2017)	Total sample of 127 participants: 41 with LBD, 47 with AD, and 39 controls	78.3 (7.1)	✓	✓	—	43 (34)/84 (66)	MMSE CAMCOG UPDRS-III NPI CAF TIV MRI CSF VBM	No significant differences were observed between AD and LBD in orientation, short- or long-term memory, recent memory, language, attention, abstract reasoning, perception, and visuoconstructive skills. Both diseases showed a similar pattern of atrophy in the hippocampus bilaterally.	
Colloby et al. ⁶⁵ (2014)	Total sample of 127 participants: 41 with LBD, 47 with AD, and 39 controls	78.3 (7.1)	✓	✓	✓	43 (34)/84 (66)	MMSE CAMCOG NPI RBD UPDRS-III CAF TIV VBM MRI	No significant differences were observed between AD and LBD in impairment of memory, orientation, language, visuoconstructive skills, or attention. From an anatomical/functional perspective, patients with AD present greater grey matter loss in the cerebellum and bilateral temporal lobe. AD is also associated with white matter loss in the bilateral middle cerebellar peduncle. Patients with LBD presented grey matter loss in the cerebellum, although this was not significant compared to AD.	

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Kraybill et al. ⁶⁶ (2005)	Total sample of 135 participants: 48 with AD, 22 with LBD, and 65 EA plus LBD	76 (6.8)	✓	✓	—	77 (57)/58 (43)	MMSE DRS TMT-A WAIS-R DS CERAD-N FOME WMS WAIS-R TMT-A TMT-B ASN	Patients with AD presented greater impairment of memory (delayed recall) and naming. LBD was associated with greater deficits in cognitive function and attention. No significant differences between groups were observed for visuoconstructive skills.
Gomperts et al. ⁶⁷ (2008)	Total sample of 78 participants: 8 with LBD, 7 with Parkinson's disease dementia, 11 with Parkinson's disease, 15 with AD, and 37 controls	71.3 (2.1)	✓	✓	—	39 (50)/39 (50)	CDR UPDRS MMSE BDS DS TMT-A TMT-B CFL PF BNT LM FRSRT FCSRT BVDT MCFS WAIS-R NPI GDS AMNART-VIQ PET PiB-PET	LBD was associated with greater impairment of attention, executive function, delayed recall, episodic memory, naming, and semantic and phonological verbal fluency, compared to AD. Patients with AD presented greater temporal lobe atrophy and less marked occipital lobe involvement. Patients with LBD presented greater involvement of the lateral parietal lobe, precuneus, and posterior cingulate cortex.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/ ♂		
			60-70	71-80	> 80	n (%)		
Taylor et al. ⁶⁸ (2013)	Total sample of 42 participants: 23 with AD and 19 with LBD	77.5 (10.0)	✓	✓	✓	19 (45)/23 (55)	MMSE CAMCOG UPDRS-III CAF	The most affected domains in AD are memory, language, orientation, and abstract reasoning. Patients with LBD presented greater impairment of attention. From an anatomical/functional perspective, both groups presented atrophy of the bilateral thalamus, putamen, and precuneus, and of the left occipital lobe. Patients with AD also presented atrophy of the bilateral pre- and post-central gyri and cerebellum, as well as the left temporal lobe. Patients with LBD displayed atrophy of the bilateral anterior cingulate cortex and caudate nucleus.
Hamilton et al. ⁶⁹ (2008)	Total sample of 66 participants: 22 with LBD and 44 with AD	72.7 (5.9)	✓	✓	—	32 (48)/34 (52)	CRT SPECT MMSE MDRS WISC-R BD BNT ADL BI	Patients with LBD presented severely impaired visuconstructive skills, whereas AD was associated with greater impairment of language abilities. These alterations were predictive of progression to severe overall cognitive impairment in both groups.
Fong et al. ⁷⁰ (2011)	Total sample of 46 participants: 14 with LBD, 16 with AD, and 16 controls	70.7 (9.5)	✓	✓	—	21 (46)/25 (54)	MMSE CT MRI CERAD COWAT BNT RAVLT SF FAS DS	Patients with LBD performed better in delayed recall, recognition, and phonological verbal fluency, compared to those with AD. Patients with AD presented greater impairment in naming and semantic verbal fluency tasks. The anatomical/functional study revealed hypoperfusion of the bilateral frontal lobe, precuneus, and parietal and occipital lobes in LBD. The AD group showed frontal, parietal, and occipital hypoperfusion; however, this alteration was more severe in LBD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Ballard et al. ⁷¹ (2001)	Total sample of 190 participants: 85 with LBD, 80 with AD, and 35 controls	77 (6.8)	✓	✓	—	103 (54)/87 (46)	MMSE CDS M-UPDRS COGDRAS-D CRT CogRT DS WAIS-R VIG	While both LBD and AD present significant general cognitive impairment, the differentiating factor is the deficit and fluctuations in attention performance, which are more severe in LBD.
Kawai et al. ⁷² (2013)	Total sample of 440 participants: 402 with AD and 38 with LBD	78.8 (6.3)	✓	✓	—	313 (71)/127 (29)	MMSE ADAS-Jcog FAB RCPM DS LM WMS-R MRI CT	Patients with AD presented greater impairment of orientation to time and space, as well as delayed recall. In turn, patients with LBD showed greater impairment of attention, executive function, visuoconstructive skills, and phonological verbal fluency. These domains are proposed as key factors for differential diagnosis of AD and LBD.
Ryman et al. ⁷³ (2021)	Total sample of 2433 participants: 111 with LBD, 741 with AD plus LBD, 1357 with AD, and 224 controls	80.1 (9.5)	✓	✓	✓	1086 (45)/1347 (55)	MMSE CERAD TMT-A TMT-B DSym VF SF BNT DS BI	LBD was associated with severe impairment of executive function and visuospatial skills. The AD group showed significant deficits in memory and language. Both groups showed similar levels of attention impairment.
Watson et al. ⁷⁴ (2015)	Total sample of 94 participants: 31 with LBD, 30 with AD, and 33 controls	77.5 (6.0)	✓	✓	—	33 (35)/61 (65)	MMSE	Patients with AD presented greater overall cognitive impairment than those with LBD. They also presented severe delays in attention, memory, orientation to space and time, language, executive function, and visuoconstructive skills.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
							CAMCOG UPDRS-III NPI B-ADL	In the AD group, the anatomical/functional study revealed atrophy of temporal and parietal areas, extending to frontal regions. These patients presented severe atrophy of the hippocampus, precuneus, and entorhinal cortex. Patients with LBD displayed atrophy in the inferior parietal region, posterior cingulate, and fusiform gyrus.
Prats-Sedano et al. ⁷⁵ (2021)	Total sample of 182 participants: 76 with LBD, 40 with AD, and 66 controls	73.8 (7.1)	✓	✓	—	48 (26)/134 (74)	MRI CSF MMSE	Patients with AD presented greater memory impairment. Patients with LBD showed greater deficits in visuospatial abilities. No significant differences were observed between groups for attention, orientation, or language. Patients with LBD performed slightly worse in verbal fluency than those with AD.
Mitolo et al. ⁷⁶ (2016)	Total sample of 228 participants: 28 with LBD, 115 with AD, and 85 controls	73.1 (8.6)	✓	✓	—	—/—	ACE-R BI MMSE MDRS VOT CDT WISC-R BD BNT SF CVLT WMS-R LM	Patients with LBD presented greater deficits in visuospatial and visuoconstructive abilities, semantic verbal fluency, and language. While language was also impaired in AD, episodic memory was most significantly affected in this group.
Kenny et al. ³² (2012)	Total sample of 47 participants: 15 with LBD, 16 with AD, and 16 controls	78.1 (7.7)	✓	✓	—	—/—	MMSE UPDRS-III GDS	Despite mild differences in scores, there was no significant difference between patients with AD and LBD in overall cognitive performance. Therefore, the results demonstrate similar levels of impairment in both groups. However, memory is severely affected in patients with AD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Stavitsky et al. ⁷⁷ (2006)	Total sample of 83 participants: 55 with AD and 28 with LBD	73.3 (7.9)	✓	✓	✓	43 (52)/40 (48)	NPI	The anatomical/functional study revealed greater connectivity in the left hippocampus, right insula, and left inferior parietal lobe in patients with AD. Patients with LBD presented greater connectivity in the right posterior cingulate cortex, limbic area (left anterior cingulate), right globus pallidus, right culmen, and right cerebellar tonsil. Both groups showed similar connectivity in the insula, thalamus, and caudate nucleus.
							CAMCOG CAF fMRI Modified	Patients with AD presented greater memory impairment, whereas those with LBD presented greater impairment of visuoconstructive skills. Furthermore, patients with AD present greater behavioural alterations and increased visual hallucinations; in LBD, these symptoms tend to remain stable over the course of the disease. No significant difference between groups was observed for language deficits.
Mak et al. ⁷⁸ (2016)	Total sample of 104 participants: 35 with LBD, 34 with AD, and 35 controls	77.7 (6.0)	✓	✓	—	38 (36)/66 (64)	MMSE DS HVLt-R IADL ADL DDS MMSE	AD was associated with greater impairment of delayed recall, information retention capacity, verbal memory, and learning, compared to LBD. Patients with LBD presented greater impairment of recent memory, and greater cognitive fluctuations than patients with AD.
							CAMCOG CAF UPDRS-III NPI HVLt BVMT MRI	Patients with LBD displayed greater atrophy of the medial hippocampus. Although hippocampal atrophy is a characteristic feature of AD, the results suggest that atrophy of the CA1 region is a key factor in differentiating AD and LBD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Park et al. ⁷⁹ (2011)	Total sample of 103 participants: 10 with LBD, 76 with AD, and 17 with Parkinson's disease dementia	70.2 (7.2)	✓	✓	—	68 (66)/35 (34)	K-MMSE CDR GDS BADL DS K-BNT ROCF COWAT ST SVLT MRI	Patients with LBD present greater impairment of executive function and attention than those with AD. Both diseases were associated with impairment of episodic memory. Language and naming were impaired in both groups, although patients with LBD presented lower scores.
Breitve et al. ⁸⁰ (2018)	Total sample of 186 participants: 119 with AD and 67 with LBD	75.6 (7.5)	✓	✓	✓	116 (62)/70 (38)	MMSE CDR CVLT-II TMT-A TMT-B COWAT BNT VOSP ST I-FP-CIT SPECT MMSE	Patients with AD performed worse than those with LBD in attention, language (semantic verbal fluency), executive function, and memory (delayed recall). However, during the follow-up period, severe attention impairment in LBD was predictive of progression and deterioration in this patient group; this was not the case in AD.
Mak et al. ⁸¹ (2015)	Total sample of 69 participants: 13 with LBD, 23 with AD, and 33 controls	76.7 (6.3)	✓	✓	✓	24 (35)/45 (65)		No significant differences were observed at baseline or during follow-up between LBD and AD in the domains memory, attention, executive function, visuoconstructive skills, perception, abstract reasoning, or verbal fluency. However, greater cognitive fluctuations were observed in LBD than in AD.

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Firbank et al. ⁸² (2007)	Total sample of 46 participants: 16 with LBD, 15 with AD, and 15 controls	75.6 (7.3)	✓	✓	—	18 (39)/28 (61)	BADL UPDRS-III NPI	The anatomical/functional study revealed a pattern of atrophy involving the left superior and middle temporal gyri, extending to the left lingual gyrus. In contrast, patients with LBD displayed frontal and parietal atrophy. Overall cognitive performance was severely impaired in LBD, with alterations in attention, orientation, memory, language (verbal fluency), visuoconstructive skills, perception, abstract reasoning, and executive function.
							CogFluct CAMCOG MRI MMSE CAMCOG UPDRS NPI	
							CDS	
							UPDRS-III CT MRI	

Table 1 (Continued)

Lead author (year)	Population	Age	Age range			Sex	Instruments /measures	Main findings
		Mean (SD)	(years)			♀/♂		
			60-70	71-80	> 80	n (%)		
Watson et al. ⁸³ (2012)	Total sample of 106 participants: 35 with LBD, 36 with AD, and 35 controls	77.8 (6.0)	✓	✓	✓	38 (36)/68 (64)	MMSE CAMCOG VF BADL NPI UPDRS-III MRI	Both AD and LBD were associated with episodic memory deficits. Furthermore, phonological verbal fluency was more severely impaired in LBD than in AD. The anatomical/functional study showed that LBD was associated with atrophy of parietal, occipital, and temporal areas, with extension to the frontal lobe. AD tended to show a more generalised pattern of atrophy, involving the same areas as LBD but with greater temporal lobe atrophy.

✓/: included; —: not reported.

ACE-R: Addenbrooke's Cognitive Examination-revised; ADAS-Jcog: Alzheimer's Disease Assessment Scale-Cognitive Component-Japanese version; ADL: Activities of Daily Living; AMNART-VIQ: American version of the National Adult Reading Test, verbal IQ; ANT: Attention Network Test; ASN: α -synuclein; BADL: Bristol Activities of Daily Living Scale; BDAE: Boston Diagnostic Aphasia Examination; BDS: Blessed Dementia Scale; BI: brain imaging; BNT: Boston Naming Test; BP: blood pressure; BPRS: Brief Psychiatric Rating Scale; BVDT: Benton Visual Form Discrimination Test; BVMT: Brief Visuospatial Memory Test; BVRT: Benton Visual Retention Test; CAF: Clinician Assessment of Fluctuation; CAMCOG: Cambridge Cognitive Examination; CASI: Cognitive Abilities Screening Instrument; CDR: Clinical Dementia Rating Scale; CDS: Cornell Depression Scale; CDT: Clock Drawing Test; CERAD: Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Battery; CERAD-N: CERAD naming subtest; CFL: category fluency; COGDRAS-D: Cognitive Drug Research Computerized Assessment System for Dementia Patients; CogFluct: Cognitive Fluctuation Scale; CogRT: cognitive reaction time; COWAT: Controlled Oral Word Association Test; CRT: choice reaction time; CSF: cerebrospinal fluid; CT: computed tomography; CVLT: California Verbal Learning Test; CVLT-II: CVLT, second edition; DAN: dorsal attention network; DCFS: Dementia Cognitive Fluctuation Scale; DDS: Dependence Scale sum; D-KEFS ST: Delis-Kaplan Executive Function Sorting Test; DMN: default mode network; DMS-48: Delayed Matching to Sample-48 items; DO80: Oral Denomination-80 items; DRS: Dementia Rating Scale; DS: digit span; DSB: digit span backward; DSST: Digit Symbol Substitution Test; DSym: digit symbol; EEG: electroencephalography; eTIV: estimated total intracranial volume; FAB: Frontal Assessment Battery; FAQ: Functional Assessment Questionnaire; FAS: FAS verbal fluency; FBI: Frontal Behavioral Inventory; FCSRT: Free and Cued Selective Reminding Test; FDG-PET: [¹⁸F]-fluorodeoxyglucose positron emission tomography; FER: facial emotion recognition; FLE: Lexical Evocation; fMRI: functional magnetic resonance imaging; FOME: Fuld Object Memory Evaluation; FPN: frontoparietal network; FPRT: Faux Pas Recognition Test; FRSRT: Free Selective Reminding Test; FSE: formal semantic evocation; FSIQ: Full Scale IQ; GDS: Geriatric Depression Scale; GDS-15: Geriatric Depression Scale-15 items; GNT: Graded Naming Test; HADS: Hospital Anxiety and Depression Scale; HDRS: Hamilton Depression Rating Scale; HVLT: Hopkins Verbal Learning Test; HVLT-R: Hopkins Verbal Learning Test-Revised; IADL: Instrumental Activities of Daily Living scale; I-FP-CIT SPECT: ioflupane single-photon emission computed tomography; K-BNT: Korean version of the Boston Naming Test; K-MMSE: Korean version of the Mini-Mental State Examination; LM: logical memory; LMS-A: Logical Memory Story A-immediate and delayed recall; MCFS: Mayo Clinic Fluctuation Scale; MDRS: Mattis Dementia Rating Scale; MDS-UPDRS: Movement Disorders Society Unified Parkinson's Disease Rating Scale motor sub-scale; MEC: Miniexamen Cognoscitivo; MFCS: Mayo Fluctuation Composite Score; Mini-SEA: Mini-Social Cognition and Emotional Assessment; mMMSE: modified Mini-Mental State Examination; MMSE: Mini-Mental State Examination; MoCA: Montreal Cognitive Assessment; MRI: magnetic resonance imaging; M-UPDRS: Modified Unified Parkinson's Disease Rating Scale; NA: neuropsychological assessment; NART: National Adult Reading Test; NPI: Neuropsychiatric Inventory; PC: Pentagon Copy; PET: positron emission tomography; PF: phonological fluency; PiB-PET: Pittsburgh compound B positron emission tomography; PIQ: Performance IQ; PPT: Pyramids and Palm Trees test; RAVLT: Rey Auditory Verbal Learning Test; RBD: REM sleep behaviour disorder; RCPM: Raven's Coloured Progressive Matrices; RDT: Rosen drawing test; RMET: Reading the Mind in the Eyes Test; ROCF: Rey-Osterrieth Complex Figure; SD: standard deviation; SF: semantic fluency; SG: symbol gesture; SPECT: single-photon emission computed tomography; SRT: Selective Reminding Test; SS-IQCODE: Shortened Spanish-Informant Questionnaire on Cognitive Decline in the Elderly; ST: Stroop Test; SVLT: Seoul Verbal Learning Test; TIV: total intracranial volume; TMT: Trail-Making Test; TT: token test; UPDRS: Unified Parkinson's Disease Rating Scale; UPDRS-III: Unified Parkinson's Disease Rating Scale Part III; VBM: voxel-based morphometry; VF: verbal fluency; VH: visual hallucinations; VIG: Tasks and Digit Vigilance; VIQ: Verbal IQ; VOSP: Visual Object and Space Perception Battery; VOT: Visual Organization Test; WAB: Western Aphasia Battery; WAIS-R: Wechsler Adult Intelligence Scale, revised; WASI: Wechsler Abbreviated Scale of Intelligence; WCST: Wisconsin Card Sorting Test; WISC-R BD: Wechsler Intelligence Test for Children-Revised, block design subtest; WMS: Wechsler Memory Scale; WMS-R: Wechsler Memory Scale-Revised.

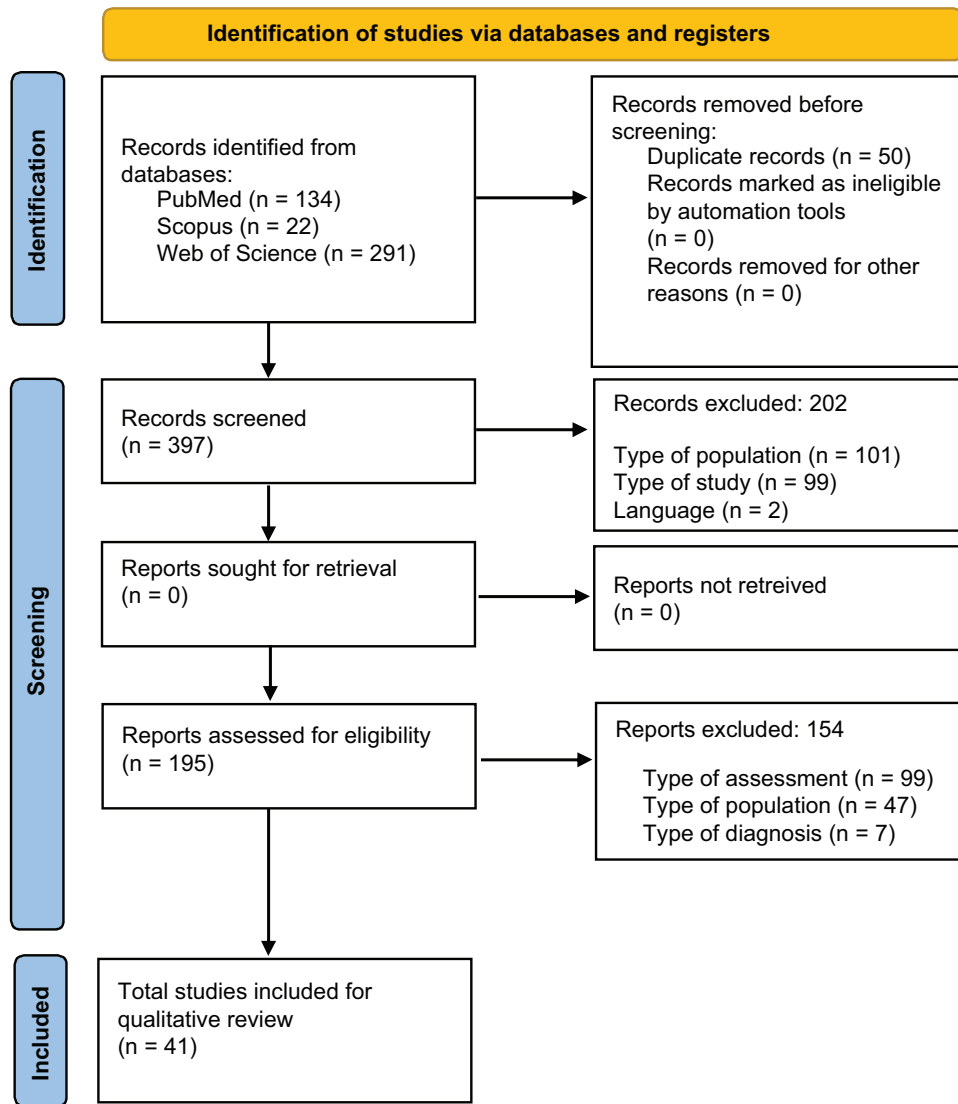


Figure 1 PRISMA flow chart.

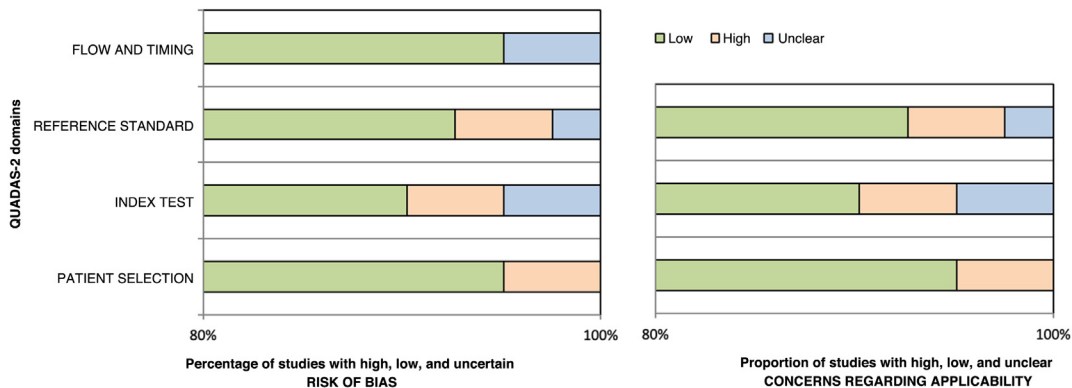


Figure 2 Assessment of risk of bias and methodological quality.

affected in LBD. Significant differences were also observed in the ability to consolidate information, with AD showing severe deficits mainly affecting delayed recall and verbal memory.⁴⁵

Another recent study⁹⁰ notes that analysis during the pre-dementia stage is key to differentiating between patients who will progress to AD or LBD. The results suggest that patients in the mild cognitive impairment stage who present

mild memory deficits tend to progress to LBD, whereas progression to AD is associated with severe memory impairment. Previous studies^{91–93} report that this performance is also observed in patients with subjective memory complaints. Similarly, Lemstra et al.⁹⁴ highlight the relevance of LBD co-occurring with AD, as memory decline tends to be severe in these patients, even at early stages. Nonetheless, the results reported in the literature are contradictory, with other studies suggesting that memory performance is better in patients with AD plus LBD than in those with pure AD^{19,95}; on the other hand, Azar et al.³⁰ suggest that memory impairment is no different in patients with AD and concomitant LBD. These discrepancies may be explained by differences in sample size. It should also be noted that there is a clear overlap in the cognitive symptoms and patterns of brain involvement between AD and AD plus LBD; as a result, differential diagnosis is relatively imprecise.^{77,96}

Attention and executive function

Attention deficit and executive function decline were identified as cognitive markers for the differential diagnosis of AD and LBD. In line with our own findings, Xu et al.⁹⁷ report that both domains are more severely impaired in LBD than in AD. Furthermore, several authors^{98–100} suggest that selective attention is a differential cognitive marker in early stages. Although attention deficit manifests early in AD, it is more evident in moderate and severe disease.⁷¹ However, other studies present contradictory results,^{101–103} suggesting that divided attention is key to differentiating between these diseases, as it is more severely impaired in LBD.

Regarding executive function, specific subdomains are proposed for differential diagnosis. For instance, the evidence suggests that inhibitory control^{97,104,105} and cognitive flexibility^{50,106–109} are affected in LBD. In contrast, Bailon et al.¹¹⁰ and Bussè et al.¹⁰³ suggest that cognitive flexibility does not present significant impairment in LBD. Similarly, discrepancies were also observed regarding performance in abstract reasoning tasks and conceptual skills. While some authors assert that these domains are unaffected,^{105,108} Perri et al.¹¹¹ report severe impairment in patients with LBD compared to those with AD.

Visuocognitive skills

Visuoconstructive and visuospatial deficits are observed in both AD and LBD. Ferman et al.¹⁰⁷ note that visuoconstructive impairment enables accurate differentiation between LBD, AD, and healthy elderly adults (sensitivity: 80%; specificity: 90%). Similarly, other studies suggest that visuoconstructive skills are severely impaired in LBD, whereas AD is associated with severe impairment of episodic memory, favouring differential diagnosis.^{112–114} Furthermore, these studies report that early stages are crucial in identifying the deficit, as motor symptoms in advanced stages of LBD contribute to diagnostic errors.

Visuospatial skills are also proposed as a key element in differential diagnosis. Classically, severe visuospatial deficit has been considered a differentiating factor in LBD^{66,77,95,98};

however, interesting findings are also reported in AD. It has been suggested that deficits in these skills may be the first cognitive manifestations of AD, with the initial stage being fundamental to distinguishing it from LBD.^{115–118}

Changes in visuoperceptual skills have been associated with these diseases; among the different types of visuocognitive ability, these skills stand out for their high complexity. Visuoperceptual skills are reported to be significantly impaired in LBD, enabling differentiation from AD at early stages.^{22,54,62,76,119,120} However, other studies report diverging results, with these deficits also having been observed in preclinical and mild AD.^{121,122}

Semantic and phonological verbal fluency

Various studies^{62,105,106,123} report that patients with LBD present severe impairment of semantic and phonological verbal fluency, as well as deficits in naming. Ralph et al.²² support this hypothesis, but note that patients with typical AD present better phonological performance. Other researchers^{124–126} argue that the deficit in semantic verbal fluency in AD is due to a loss of semantic knowledge. Similarly, patients with LBD also show language impairment, but this differs from that observed in AD in that lexical and semantic abilities are spared.¹²⁷ Additionally, it should be noted that verbal fluency tests involving verbs have also been proposed as cognitive/linguistic markers. For instance, Delbeuck et al.¹²⁸ found that patients with LBD were able to produce higher numbers of verbs than those with AD.

Neuropsychological assessment tools

The studies reviewed frequently used the MMSE and MoCA test to evaluate cognitive performance. The MMSE is a brief, simple cognitive screening test, but is unable to exhaustively assess the different cognitive domains. However, its use is encouraged in clinical practice as it is able to detect progression of cognitive impairment and to differentiate AD from LBD. In fact, patients with LBD show significant reductions in scores at annual follow-up visits compared to those with AD.^{84,88,129} Similarly, longitudinal studies^{80,130,131} highlight the rapid decline observed in MMSE scores in LBD. Although the MMSE is more useful than the MoCA test for identifying cognitive impairment in LBD,¹³² it is not recommended for studying progression of the diagnosis and cognitive decline in longitudinal studies.¹³³ Using the MMSE to monitor disease progression is only helpful in more advanced stages of LBD (General Deterioration Scale scores > 4), as results are often normal in earlier stages, when abnormal results indicate an amnesic profile suggesting copresence of AD. Furthermore, variation is reported in MMSE scores, which differ significantly in LBD (lower scores on the MoCA test and higher scores on the MMSE).¹³⁴

Our results showed that the Free and Cued Selective Reminding Test is not frequently used, despite its demonstrated efficacy in identifying patients with AD.¹³⁵ Bussè et al.¹³⁶ studied the test's effectiveness for differentiating LBD from AD, finding that although patients with LBD pre-

sented greater overall performance, total recall and delayed free recall were key to differentiating the 2 diseases.

Language evaluation is mainly based on such tests as the BNT, FAS, semantic verbal fluency, and the language subdomains of the CAMCOG. Although these tools offer a general view of language performance, they do not assess specific subdomains in detail. For instance, the Mini–Linguistic State Examination (MLSE),¹³⁷ recently validated for classifying primary progressive aphasia in Spanish-speaking populations,¹³⁸ may be useful for better characterising language deficits in AD and LBD based on phonology, semantics, and syntax (including working memory). However, language evaluation should also consider spontaneous language, which is thought to play a key role in differentiating between these 2 diseases.¹³⁹

Neuroimaging techniques and atrophy pattern

Structural MRI is frequently used to diagnose AD and LBD, and is able to identify structural damage and grey matter loss. Previous studies^{140–142} confirm these findings, highlighting the overlap between the patterns of atrophy in LBD and in AD. However, atrophy is more diffuse in LBD, with moderate preservation of the medial temporal lobe. Similarly, several other studies^{143–145} report that atrophy in AD involves not only the temporal lobe, but also the cingulate, parahippocampal regions, and the precuneus. In LBD, cortical thinning is reported to affect superior temporo-occipital regions, the lateral orbitofrontal cortex, and the posterior and medial cingulate cortex.^{143–145}

Furthermore, fMRI studies show greater connectivity in the inferior parietal cortex and putamen in LBD, whereas patients with AD display greater connectivity in frontoparietal areas, the medial prefrontal cortex, the primary visual cortex, and the left hippocampus.^{146,147} In contrast, Schumacher et al.⁴⁷ recommend caution in the interpretation of fMRI results, as the increase in frontotemporal connectivity in LBD overlaps with that reported in AD.

The growing interest in establishing an accurate differential diagnosis between AD and LBD has promoted the use of PET studies. This technique is useful for identifying AD and for distinguishing it from LBD, with the hippocampus being the region of interest. Tau-PET reveals greater concentration of tau protein in the hippocampus in AD, with more moderate concentrations in LBD.^{148–150} Furthermore, FDG-PET results show hypometabolism in the occipital region in LBD, with sparing of the posterior cingulate, whereas hypometabolism in AD affects temporo-occipital regions.⁵ Finally, Marquié et al.¹⁵¹ suggest that the radiotracer ¹⁸F-AV-1451 in tau-PET studies may be a potential biomarker for differential diagnosis, as it presents high affinity for tau neurofibrillary pathology in AD.

Limitations

This study presents 4 significant limitations. Firstly, the review does not include a meta-analysis, which would improve the interpretation of the neuropsychological and

imaging findings in AD and LBD. Secondly, only 3 databases were consulted (PubMed, Scopus, and Web of Science). Future studies should search a larger number of databases to obtain a global view of the differential diagnosis of these diseases. Thirdly, the search strategy was limited, focusing only on observational studies of diagnostic tests and excluding other study designs and even neuropathological diagnosis. Similarly, given the range of years of publication, we consider that there is a need for future studies in this area to take into account updates to diagnostic criteria. Finally, we excluded atypical AD profiles, such as primary progressive aphasia and posterior cortical atrophy, as numerous studies suggest that these share symptoms with LBD.^{152–155}

Contributions to clinical practice

To our knowledge, this is the first systematic review addressing differential diagnosis between AD and LBD that considers both the neuropsychological profile and the anatomical/functional correlates of each disease. These findings are valuable beyond the research setting, contributing directly to improving screening and accurate differential diagnosis in the early stages of both diseases.

Future lines of research

Future studies should consider using fMRI to comprehensively analyse semantic and phonological processing and to analyse connectivity patterns in brain areas responsible for language processing in AD and LBD. A final future challenge will be to promote studies that expand clinical diagnosis beyond neuropsychological symptoms and the pattern of atrophy, seeking to characterise the complete neuropsychiatric picture of each disease. Regarding the characteristics of psychoemotional interaction and social cognition, there is a need for deeper understanding of abilities related to the theory of mind, as few studies have examined the relationship between these processes and neuropsychiatric symptoms in early stages of dementia.^{156,157}

Conclusion

This review identified the main similarities and differences in neuropsychological symptoms and patterns of cortical atrophy between patients with AD and LBD. Although the domains of memory, attention, executive function, visuospatial/visuoconstructive skills, and semantic and phonological verbal fluency are proposed as important cognitive markers, the level of severity at early stages seems to be key to the differential diagnosis of these 2 diseases. The pattern of fronto-parieto-temporal, hippocampal, and precuneus atrophy tends to overlap in both diseases. In this regard, the identification of functional alterations using fMRI would play an important role in early and preclinical stages of both neurodegenerative diseases.

Finally, language is increasingly recognised as another potential marker supporting differential diagnosis in AD. Nonetheless, assessment of spontaneous language should be promoted as a complementary resource for early, accurate differentiation between AD and LBD.

Funding

This study has received no specific funding from any public, private, or non-profit organisation.

Declaration of competing interest

None.

Acknowledgements

David Toloza-Ramirez thanks the Chilean National Agency for Research and Development (ANID)-Subdirección de Capital Humano/Doctorado Nacional/2021-21212181, and Teresa Julio-Ramos thanks ANID-Subdirección de Capital Humano/Doctorado Nacional/2023-21230591. The authors also thank Daniela Aguirre of the neuroscience masters programme at Universidad Autónoma de Chile, Temuco campus, for her support in the preliminary theoretical construction of this study.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi: <https://doi.org/10.1016/j.nrleng.2025.10.001>.

References

- [1]. Chin KS, Teodorczuk A, Watson R. Dementia with Lewy bodies: challenges in the diagnosis and management. *Australian New Zealand J Psychiatry*. 2019;53(4):291–303, <http://dx.doi.org/10.1177/0004867419835029>.
- [2]. Leng F, Edison P. Neuroinflammation and microglial activation in Alzheimer disease: where do we go from here? *Nat Rev Neurol*. 2021;17(3):157–72, <http://dx.doi.org/10.1038/s41582-020-00435-y>.
- [3]. Orad RI, Shiner T. Differentiating dementia with Lewy bodies from Alzheimer's disease and Parkinson's disease dementia: an update on imaging modalities. *J Neurol*. 2022;269(2):639–53, <http://dx.doi.org/10.1007/s00415-021-10402-2>.
- [4]. Sperling RA, Aisen PS, Beckett LA, Bennett DA, Craft S, Fagan AM, et al. Toward defining the preclinical stages of Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's Dementia*. 2011;7(3):280–92, <http://dx.doi.org/10.1016/j.jalz.2011.03.003>.
- [5]. Burkett BJ, Babcock JC, Lowe VJ, Graff-Radford J, Subramaniam RM, Johnson DR. PET imaging of dementia. *Clin Nucl Med*. 2022;47(9):763–73, <http://dx.doi.org/10.1097/rlu.0000000000004251>.
- [6]. Serrano-Pozo A, Frosch MP, Masliah E, Hyman BT. Neuropathological Alterations in Alzheimer Disease. *Csh Perspect Med*. 2011;1(1):a006189, <http://dx.doi.org/10.1101/cshperspect.a006189>.
- [7]. Kreisl WC, Lyoo CH, Liow J-S, Snow J, Page E, Jenko KJ, et al. Distinct patterns of increased translocator protein in posterior cortical atrophy and amnesic Alzheimer's disease. *Neurobiol Aging*. 2017;51:132–40, <http://dx.doi.org/10.1016/j.neurobiolaging.2016.12.006>.
- [8]. Hansson O, Lehmann S, Otto M, Zetterberg H, Lewczuk P. Advantages and disadvantages of the use of the CSF Amyloid β (A β) 42/40 ratio in the diagnosis of Alzheimer's Disease. *Alzheimer's Res Ther*. 2019;11(1):34, <http://dx.doi.org/10.1186/s13195-019-0485-0>.
- [9]. Jellinger KA, Korczyn AD. Are dementia with Lewy bodies and Parkinson's disease dementia the same disease? *Bmc Med*. 2018;16(1):34, <http://dx.doi.org/10.1186/s12916-018-1016-8>.
- [10]. Verry M, Duyckaerts C. Cognitive deficit, and neuropathological correlates, in the oldest-old. *Rev Neurol*. 2020;176(9):670–6, <http://dx.doi.org/10.1016/j.neurol.2020.01.355>.
- [11]. Kindell J, Keady J, Sage K, Wilkinson R. Everyday conversation in dementia: a review of the literature to inform research and practice. *Int J Lang Amp Commun Disord*. 2017;52(4):392–406, <http://dx.doi.org/10.1111/1460-6984.12298>.
- [12]. Toloza-Ramírez D, Méndez-Orellana C, Martella D. Diagnóstico neuropsicológico diferencial en enfermedad de Alzheimer y demencia frontotemporal: una revisión sistemática cualitativa. *Neurol Perspect*. 2021;1(1):82–97, <http://dx.doi.org/10.1016/j.neurop.2021.03.004>.
- [13]. McKeith IG, Boeve BF, Dickson DW, Halliday G, Taylor J-P, Weintraub D, et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. *Neurology*. 2017;89(1):88–100, <http://dx.doi.org/10.1212/wnl.0000000000004058>.
- [14]. Hogan DB, Fiest KM, Roberts JI, Maxwell CJ, Dykeman J, Pringsheim T, et al. The prevalence and incidence of dementia with Lewy bodies: a systematic review. *Can J Neurological Sci*. 2016;43(S1):S83–95, <http://dx.doi.org/10.1017/cjn.2016.2>.
- [15]. Kosaka K. Latest concept of Lewy body disease. *Psychiat Clin Neuros*. 2014;68(6):391–4, <http://dx.doi.org/10.1111/pcn.12179>.
- [16]. Spillantini MG, Schmidt ML, Lee VM-Y, Trojanowski JQ, Jakes R, Goedert M. α -Synuclein in Lewy bodies. *Nature*. 1997;388(6645):839–40, <http://dx.doi.org/10.1038/42166>.
- [17]. Magdalino NK, Golden HL, Nicholas JM, Witoonpanich P, Mummery CJ, Morris HR, et al. Verbal adynamia in parkinsonian syndromes: behavioral correlates and neuroanatomical substrate. *Neurocase*. 2018;24(4):204–12, <http://dx.doi.org/10.1080/13554794.2018.1527368>.
- [18]. Metzler-Baddeley C. A review of cognitive impairments in dementia with lewy bodies relative to Alzheimer's disease and Parkinson's disease with dementia. *Cortex*. 2007;43(5):583–600, [http://dx.doi.org/10.1016/s0010-9452\(08\)70489-1](http://dx.doi.org/10.1016/s0010-9452(08)70489-1).
- [19]. Yoshizawa H, Vonsattel JPG, Honig LS. Early neuropsychological discriminants for Lewy body disease: an autopsy series. *J Neurology Neurosurg Psychiatry*. 2013;84(12):1326, <http://dx.doi.org/10.1136/jnnp-2012-304381>.

- [20]. Calderon J, Perry RJ, Erzinclioglu SW, Berrios GE, Denning TR, Hodges JR. Perception, attention, and working memory are disproportionately impaired in dementia with Lewy bodies compared with Alzheimer's disease. *J Neurol Neurosurg Psychiatry*. 2001;70(2):157, <http://dx.doi.org/10.1136/jnnp.70.2.157>.
- [21]. Collerton D, Burn D, McKeith I, O'Brien J. Systematic review and meta-analysis show that dementia with Lewy bodies is a visual-perceptual and attentional-executive dementia. *Dement Geriatr Cogn*. 2003;16(4):229–37, <http://dx.doi.org/10.1159/000072807>.
- [22]. Ralph MAL, Powell J, Howard D, Whitworth AB, Garrard P, Hodges JR. Semantic memory is impaired in both dementia with Lewy bodies and dementia of Alzheimer's type: a comparative neuropsychological study and literature review. *J Neurol Neurosurg Psychiatry*. 2001;70(2):149, <http://dx.doi.org/10.1136/jnnp.70.2.149>.
- [23]. Ala TA, Yang KH, Sung JH, Frey WH. Hallucinations and signs of parkinsonism help distinguish patients with dementia and cortical Lewy bodies from patients with Alzheimer's disease at presentation: a clinicopathological study. *J Neurol Neurosurg Psychiatry*. 1997;62(1):16, <http://dx.doi.org/10.1136/jnnp.62.1.16>.
- [24]. Merdes AR, Hansen LA, Jeste DV, Galasko D, Hofstetter CR, Ho GJ, et al. Influence of Alzheimer pathology on clinical diagnostic accuracy in dementia with Lewy bodies. *Neurology*. 2003;60(10):1586–90, <http://dx.doi.org/10.1212/01.wnl.0000065889.42856.f2>.
- [25]. Lopez OL, Hamilton RL, Becker JT, Wisniewski S, Kaufer DI, DeKosky ST. Severity of cognitive impairment and the clinical diagnosis of AD with Lewy bodies. *Neurology*. 2000;54(9):1780–7, <http://dx.doi.org/10.1212/wnl.54.9.1780>.
- [26]. Hohl U, Tiraboschi P, Hansen LA, Thal LJ, Corey-Bloom J. Diagnostic accuracy of dementia with Lewy bodies. *Arch Neurol-chicago*. 2000;57(3):347, <http://dx.doi.org/10.1001/archneur.57.3.347>.
- [27]. Irwin DJ, Hurtig HI. The contribution of tau, amyloid-beta and alpha-synuclein pathology to dementia in Lewy body disorders. *J Alzheimer's Dis Park*. 2018;8(4):1–8, <http://dx.doi.org/10.4172/2161-0460.1000444>.
- [28]. Oda H, Yamamoto Y, Maeda K. Neuropsychological profile of dementia with Lewy bodies. *Psychogeriatrics*. 2009;9(2):85–90, <http://dx.doi.org/10.1111/j.1479-8301.2009.00283.x>.
- [29]. Ardle RM, Galna B, Donaghy P, Thomas A, Rochester L. Do Alzheimer's and Lewy body disease have discrete pathological signatures of gait? *Alzheimer's Dementia*. 2019;15(10):1367–77, <http://dx.doi.org/10.1016/j.jalz.2019.06.4953>.
- [30]. Azar M, Chapman S, Gu Y, Leverenz JB, Stern Y, Cosentino S. Cognitive tests aid in clinical differentiation of Alzheimer's disease versus Alzheimer's disease with Lewy body disease: evidence from a pathological study. *Alzheimer's Dementia*. 2020;16(8):1173–81, <http://dx.doi.org/10.1002/alz.12120>.
- [31]. O'Brien JT, Firbank MJ, Davison C, Barnett N, Bamford C, Donaldson C, et al. 18F-FDG PET and perfusion SPECT in the diagnosis of Alzheimer and Lewy body dementias. *J Nucl Med*. 2014;55(12):1959–65, <http://dx.doi.org/10.2967/jnumed.114.143347>.
- [32]. Kenny ER, Blamire AM, Firbank MJ, O'Brien JT. Functional connectivity in cortical regions in dementia with Lewy bodies and Alzheimer's disease. *Brain*. 2012;135(2):569–81, <http://dx.doi.org/10.1093/brain/awr327>.
- [33]. Noe E, Marder K, Bell KL, Jacobs DM, Manly JJ, Stern Y. Comparison of dementia with Lewy bodies to Alzheimer's disease and Parkinson's disease with dementia. *Mov Disord*. 2004;19(1):60–7, <http://dx.doi.org/10.1002/mds.10633>.
- [34]. Aarsland D, Ballard C, Larsen JP, McKeith I. A comparative study of psychiatric symptoms in dementia with Lewy bodies and Parkinson's disease with and without dementia. *Int J Geriatr Psychiatry*. 2001;16(5):528–36, <http://dx.doi.org/10.1002/gps.389>.
- [35]. Chiu P-Y, Tsai C-T, Chen P-K, Chen W-J, Lai T-J. Neuropsychiatric symptoms in Parkinson's disease dementia are more similar to Alzheimer's disease than dementia with Lewy bodies: a case-control study. *PLoS One*. 2016;11(4):e0153989, <http://dx.doi.org/10.1371/journal.pone.0153989>.
- [36]. Bougea A, Stefanis L, Paraskevas GP, Emmanouilidou E, Efthymiopoulou E, Vekrelis K, et al. Neuropsychiatric symptoms and α -Synuclein profile of patients with Parkinson's disease dementia, dementia with Lewy bodies and Alzheimer's disease. *J Neurol*. 2018;265(10):2295–301, <http://dx.doi.org/10.1007/s00415-018-8992-7>.
- [37]. Graff-Radford J, Yong KXX, Apostolova LG, Bouwman FH, Carrillo M, Dickerson BC, et al. New insights into atypical Alzheimer's disease in the era of biomarkers. *Lancet Neurol*. 2021;20(3):222–34, [http://dx.doi.org/10.1016/s1474-4422\(20\)30440-3](http://dx.doi.org/10.1016/s1474-4422(20)30440-3).
- [38]. Scarmeas N, Brandt J, Blacker D, Albert M, Hadjigeorgiou G, Dubois B, et al. Disruptive behavior as a predictor in Alzheimer disease. *Arch Neurol*. 2007;64(12):1755–61, <http://dx.doi.org/10.1001/archneur.64.12.1755>.
- [39]. Scarmeas N, Brandt J, Albert M, Hadjigeorgiou G, Papadimitriou A, Dubois B, et al. Delusions and hallucinations are associated with worse outcome in Alzheimer disease. *Arch Neurol*. 2005;62(10):1601–8, <http://dx.doi.org/10.1001/archneur.62.10.1601>.
- [40]. Devanand DP, Lee S, Huey ED, Goldberg TE. Associations between neuropsychiatric symptoms and neuropathological diagnoses of Alzheimer disease and related dementias. *Jama Psychiat*. 2022;79(4):359–67, <http://dx.doi.org/10.1001/jamapsychiatry.2021.4363>.
- [41]. de Oliveira FF, Miraldo MC, de Castro-Neto EF, de Almeida SS, Matas SLA, Bertolucci PHF, et al. Associations of Neuropsychiatric features with cerebrospinal fluid biomarkers of amyloidogenesis and neurodegeneration in dementia with Lewy bodies compared with Alzheimer's disease and cognitively healthy people. *J Alzheimer's Dis*. 2021;81(3):1295–309, <http://dx.doi.org/10.3233/jad-210272>.
- [42]. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71, <http://dx.doi.org/10.1136/bmj.n71>.
- [43]. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan—a web and mobile app for systematic reviews. *Syst Rev*. 2016;5(1):210, <http://dx.doi.org/10.1186/s13643-016-0384-4>.
- [44]. Whiting PF, Rutjes AWS, Westwood ME, Mallett S, Deeks JJ, Reitsma JB, et al. QUADAS-2: a revised tool for the quality assessment of diagnostic accuracy studies. *Ann Intern Med*. 2011;155(8):529, <http://dx.doi.org/10.7326/0003-4819-155-8-201110180-00009>.
- [45]. Buciu M, Whitwell JL, Kananuki K, Graff-Radford J, Machulda MM, Duffy JR, et al. Lewy body disease is a contributor to logopenic progressive aphasia phenotype. *Ann Neurol*. 2021;89(3):520–33, <http://dx.doi.org/10.1002/ana.25979>.

- [46]. Tak K, Lee S, Choi E, Suh SW, Oh DJ, Moon W, et al. Magnetic resonance imaging texture of medial Pulvinar in dementia with Lewy bodies. *Dement Geriatr Cogn*. 2020;49(1):8–15, <http://dx.doi.org/10.1159/000506798>.
- [47]. Schumacher J, Peraza LR, Firbank M, Thomas AJ, Kaiser M, Gallagher P, et al. Functional connectivity in dementia with Lewy bodies: a within- and between-network analysis. *Hum Brain Mapp*. 2018;39(3):1118–29, <http://dx.doi.org/10.1002/hbm.23901>.
- [48]. Donaghy PC, Firbank MJ, Thomas AJ, Lloyd J, Petrides G, Barnett N, et al. Clinical and imaging correlates of amyloid deposition in dementia with Lewy bodies. *Mov Disord*. 2018;33(7):1130–8, <http://dx.doi.org/10.1002/mds.27403>.
- [49]. Elder GJ, Mactier K, Colloby SJ, Watson R, Blamire AM, O'Brien JT, et al. The influence of hippocampal atrophy on the cognitive phenotype of dementia with Lewy bodies. *Int J Geriatr Psychiatry*. 2017;32(11):1182–9, <http://dx.doi.org/10.1002/gps.4719>.
- [50]. Kemp J, Philippi N, Philipps C, Demuynck C, Albasser T, Martin-Hunyadi C, et al. Cognitive profile in prodromal dementia with Lewy bodies. *Alzheimer's Res Ther*. 2017;9(1):19, <http://dx.doi.org/10.1186/s13195-017-0242-1>.
- [51]. Peraza LR, Colloby SJ, Deboys L, O'Brien JT, Kaiser M, Taylor J-P. Regional functional synchronizations in dementia with Lewy bodies and Alzheimer's disease. *Int Psychogeriatr*. 2016;28(7):1143–51, <http://dx.doi.org/10.1017/s1041610216000429>.
- [52]. Kobeleva X, Firbank M, Peraza L, Gallagher P, Thomas A, Burn DJ, et al. Divergent functional connectivity during attentional processing in Lewy body dementia and Alzheimer's disease. *Cortex*. 2017;92:8–18, <http://dx.doi.org/10.1016/j.cortex.2017.02.016>.
- [53]. Schumacher J, Cromarty R, Gallagher P, Firbank MJ, Thomas AJ, Kaiser M, et al. Structural correlates of attention dysfunction in Lewy body dementia and Alzheimer's disease: an ex-Gaussian analysis. *J Neurol*. 2019;266(7):1716–26, <http://dx.doi.org/10.1007/s00415-019-09323-y>.
- [54]. Heitz C, Noblet V, Philipps C, Cretin B, Vogt N, Philippi N, et al. Cognitive and affective theory of mind in dementia with Lewy bodies and Alzheimer's disease. *Alzheimer's Res Ther*. 2016;8(1):10, <http://dx.doi.org/10.1186/s13195-016-0179-9>.
- [55]. Chabran E, Noblet V, Sousa PLde, Demuynck C, Philippi N, Mutter C, et al. Changes in gray matter volume and functional connectivity in dementia with Lewy bodies compared to Alzheimer's disease and normal aging: implications for fluctuations. *Alzheimer's Res Ther*. 2020;12(1):9, <http://dx.doi.org/10.1186/s13195-019-0575-z>.
- [56]. Yamamoto E, Mourany L, Colleran R, Whitman C, Touse B. Utility of Montreal cognitive assessment in differentiating dementia with Lewy bodies from Alzheimer's dementia. *Am J Alzheimers Dis Other Dement*. 2017;32(8):468–71, <http://dx.doi.org/10.1177/1533317517725811>.
- [57]. Firbank M, Kobeleva X, Cherry G, Killen A, Gallagher P, Burn DJ, et al. Neural correlates of attention-executive dysfunction in Lewy body dementia and Alzheimer's disease. *Hum Brain Mapp*. 2016;37(3):1254–70, <http://dx.doi.org/10.1002/hbm.23100>.
- [58]. Peraza LR, Taylor J-P, Kaiser M. Divergent brain functional network alterations in dementia with Lewy bodies and Alzheimer's disease. *Neurobiol Aging*. 2015;36(9):2458–67, <http://dx.doi.org/10.1016/j.neurobiolaging.2015.05.015>.
- [59]. Brenowitz WD, Hubbard RA, Keene CD, Hawes SE, Longstreth WT, Woltjer RL, et al. Mixed neuropathologies and associations with domain-specific cognitive decline. *Neurology*. 2017;89(17):1773–81, <http://dx.doi.org/10.1212/wnl.0000000000004567>.
- [60]. Pizzi SD, Franciotti R, Taylor J-P, Thomas A, Tartaro A, Onofri M, et al. Thalamic involvement in fluctuating cognition in dementia with Lewy bodies: magnetic resonance evidences. *Cereb Cortex*. 2015;25(10):3682–9, <http://dx.doi.org/10.1093/cercor/bhu220>.
- [61]. Jiménez-Huete A, Riva E, Toledano R, Campo P, Esteban J, Barrio AD, et al. Differential diagnosis of degenerative dementias using basic neuropsychological tests. *Am J Alzheimers Dis Other Dement*. 2014;29(8):723–31, <http://dx.doi.org/10.1177/1533317514534954>.
- [62]. Nervi A, Reitz C, Tang M-X, Santana V, Piriz A, Reyes-Dumeyer D, et al. Comparison of clinical manifestations in Alzheimer disease and dementia with Lewy bodies. *Arch Neurol*. 2008;65(12):1634–9, <http://dx.doi.org/10.1001/archneur.65.12.1634>.
- [63]. Bradshaw J, Saling M, Hopwood M, Anderson V, Brodtmann A. Fluctuating cognition in dementia with Lewy bodies and Alzheimer's disease is qualitatively distinct. *J Neurol Neurosurg Psychiatry*. 2004;75(3):382, <http://dx.doi.org/10.1136/jnnp.2002.002576>.
- [64]. Colloby SJ, Elder GJ, Rabee R, O'Brien JT, Taylor J. Structural grey matter changes in the substantia innominata in Alzheimer's disease and dementia with Lewy bodies: a DARTEL-VBM study. *Int J Geriatr Psychiatry*. 2017;32(6):615–23, <http://dx.doi.org/10.1002/gps.4500>.
- [65]. Colloby Sean J, O'Brien John T, Taylor J-P. Patterns of cerebellar volume loss in dementia with Lewy bodies and Alzheimer's disease: a VBM-DARTEL study. *Psychiatry Res Neuroimaging*. 2014;223(3):187–91, <http://dx.doi.org/10.1016/j.pscychresns.2014.06.006>.
- [66]. Kraybill ML, Larson EB, Tsuang DW, Teri L, McCormick WC, Bowen JD, et al. Cognitive differences in dementia patients with autopsy-verified AD, Lewy body pathology, or both. *Neurology*. 2005;64(12):2069–73, <http://dx.doi.org/10.1212/01.wnl.0000165987.89198.65>.
- [67]. Gomperts SN, Rentz DM, Moran E, Becker JA, Locascio JJ, Klunk WE, et al. Imaging amyloid deposition in Lewy body diseases SYMBOL. *Neurology*. 2008;71(12):903–10, <http://dx.doi.org/10.1212/01.wnl.0000326146.60732.d6>.
- [68]. Taylor J-P, Colloby SJ, McKeith IG, O'Brien JT. Covariant perfusion patterns provide clues to the origin of cognitive fluctuations and attentional dysfunction in Dementia with Lewy bodies. *Int Psychogeriatr*. 2013;25(12):1917–28, <http://dx.doi.org/10.1017/s1041610213001488>.
- [69]. Hamilton JM, Salmon DP, Galasko D, Raman R, Emond J, Hansen LA, et al. Visuospatial deficits predict rate of cognitive decline in autopsy-verified dementia with Lewy bodies. *Neuropsychology*. 2008;22(6):729–37, <http://dx.doi.org/10.1037/a0012949>.
- [70]. Fong TG, Inouye SK, Dai W, Press DZ, Alsop DC. Association cortex hypoperfusion in mild dementia with Lewy bodies: a potential indicator of cholinergic dysfunction? *Brain Imaging Behav*. 2011;5(1):25–35, <http://dx.doi.org/10.1007/s11682-010-9108-x>.
- [71]. Ballard C, O'Brien J, Gray A, Cormack F, Ayre G, Rowan E, et al. Attention and fluctuating attention in patients with dementia with Lewy bodies and Alzheimer disease. *Arch Neurol*. 2001;58(6):977–82, <http://dx.doi.org/10.1001/archneur.58.6.977>.
- [72]. Kawai Y, Miura R, Tsujimoto M, Sakurai T, Yamaoka A, Takeda A, et al. Differentiation between AD and DLB. *Psychogeriatrics*. 2013;13(3):157–63, <http://dx.doi.org/10.1111/psyg.12019>.
- [73]. Ryman SG, Yutsis M, Tian L, Henderson VW, Montine TJ, Salmon DP, et al. Cognition at each stage of Lewy body disease with co-occurring Alzheimer's disease pathology. *J Alzheimer's Dis*. 2021;80(3):1243–56, <http://dx.doi.org/10.3233/jad-201187>.

- [74]. Watson R, Colloby SJ, Blamire AM, O'Brien JT. Assessment of regional gray matter loss in dementia with Lewy bodies: a surface-based MRI analysis. *Am J Geriatric Psychiatry*. 2015;23(1):38–46, <http://dx.doi.org/10.1016/j.jagp.2014.07.005>.
- [75]. Prats-Sedano MA, Savulich G, Surendranathan A, Donaghy PC, Thomas AJ, Rowe JB, et al. The revised Addenbrooke's Cognitive Examination can facilitate differentiation of dementia with Lewy bodies from Alzheimer's disease. *Int J Geriatr Psychiatry*. 2021;36(6):831–8, <http://dx.doi.org/10.1002/gps.5483>.
- [76]. Mitolo M, Hamilton JM, Landy KM, Hansen LA, Galasko D, Pazzaglia F, et al. Visual perceptual organization ability in autopsy-verified dementia with Lewy bodies and Alzheimer's disease. *J Int Neuropsych Soc*. 2016;22(6):609–19, <http://dx.doi.org/10.1017/s1355617716000436>.
- [77]. Stavitsky K, Brickman AM, Scarmeas N, Torgan RL, Tang M-X, Albert M, et al. The progression of cognition, psychiatric symptoms, and functional abilities in dementia with Lewy bodies and Alzheimer disease. *Arch Neurol*. 2006;63(10):1450–6, <http://dx.doi.org/10.1001/archneur.63.10.1450>.
- [78]. Mak E, Su L, Williams GB, Watson R, Firbank M, Blamire A, et al. Differential atrophy of hippocampal subfields: a comparative study of dementia with Lewy bodies and Alzheimer disease. *Am J Geriatric Psychiatry*. 2016;24(2):136–43, <http://dx.doi.org/10.1016/j.jagp.2015.06.006>.
- [79]. Park KW, Kim HS, Cheon S-M, Cha J-K, Kim S-H, Kim JW. Dementia with Lewy bodies versus Alzheimer's disease and Parkinson's disease dementia: a comparison of cognitive profiles. *J Clin Neurol*. 2010;7(1):19–24, <http://dx.doi.org/10.3988/jcn.2011.7.1.19>.
- [80]. Breivite MH, Chwiczczuk LJ, Brønnick K, Hynninen MJ, Auestad BH, Aarsland D, et al. A longitudinal study of neurocognition in dementia with Lewy bodies compared to Alzheimer's disease. *Front Neurol*. 2018;9:124, <http://dx.doi.org/10.3389/fneur.2018.00124>.
- [81]. Mak E, Su L, Williams GB, Watson R, Firbank MJ, Blamire AM, et al. Progressive cortical thinning and subcortical atrophy in dementia with Lewy bodies and Alzheimer's disease. *Neurobiol Aging*. 2015;36(4):1743–50, <http://dx.doi.org/10.1016/j.neurobiolaging.2014.12.038>.
- [82]. Firbank MJ, Blamire AM, Krishnan MS, Teodorczuk A, English P, Gholkar A, et al. Diffusion tensor imaging in dementia with Lewy bodies and Alzheimer's disease. *Psychiatry Res Neuroimaging*. 2007;155(2):135–45, <http://dx.doi.org/10.1016/j.pscychresns.2007.01.001>.
- [83]. Watson R, Blamire AM, Colloby SJ, Wood JS, Barber R, He J, et al. Characterizing dementia with Lewy bodies by means of diffusion tensor imaging. *Neurology*. 2012;79(9):906–14, <http://dx.doi.org/10.1212/wnl.0b013e318266fc51>.
- [84]. Rongve A, Soennesyn H, Skogseth R, Oesterhus R, Hortobágyi T, Ballard C, et al. Cognitive decline in dementia with Lewy bodies: a 5-year prospective cohort study. *BMJ Open*. 2016;6(2):e010357, <http://dx.doi.org/10.1136/bmjopen-2015-010357>.
- [85]. Price A, Farooq R, Yuan J-M, Menon VB, Cardinal RN, O'Brien JT. Mortality in dementia with Lewy bodies compared with Alzheimer's dementia: a retrospective naturalistic cohort study. *BMJ Open*. 2017;7(11):e017504, <http://dx.doi.org/10.1136/bmjopen-2017-017504>.
- [86]. Olchney JM, Galasko D, Salmon DP, Hofstetter CR, Hansen LA, Katzman R, et al. Cognitive decline is faster in Lewy body variant than in Alzheimer's disease. *Neurology*. 1998;51(2):351–7, <http://dx.doi.org/10.1212/wnl.51.2.351>.
- [87]. Mueller C, Soysal P, Rongve A, Isik AT, Thompson T, Maggi S, et al. Survival time and differences between dementia with Lewy bodies and Alzheimer's disease following diagnosis: a meta-analysis of longitudinal studies. *Ageing Res Rev*. 2019;50:72–80, <http://dx.doi.org/10.1016/j.arr.2019.01.005>.
- [88]. Gil LM, Aarsland D. Greater variability in cognitive decline in Lewy body dementia compared to Alzheimer's disease. *J Alzheimer's Dis*. 2020;73(4):1321–30, <http://dx.doi.org/10.3233/jad-190731>.
- [89]. Sadiq D, Whitfield T, Lee L, Stevens T, Costafreda S, Walker Z. Prodromal dementia with Lewy bodies and prodromal Alzheimer's disease: a comparison of the cognitive and clinical profiles. *J Alzheimer's Dis*. 2017;58(2):463–70, <http://dx.doi.org/10.3233/jad-161089>.
- [90]. Payne S, Shofer JB, Shutes-David A, Li G, Jankowski A, Dean P, et al. Correlates of conversion from mild cognitive impairment to dementia with Lewy bodies: data from the national Alzheimer's coordinating center. *J Alzheimer's Dis*. 2022;86(4):1643–54, <http://dx.doi.org/10.3233/jad-215428>.
- [91]. Mattsson N, Zetterberg H, Hansson O, Andreassen N, Parnetti L, Jonsson M, et al. CSF biomarkers and incipient Alzheimer disease in patients with mild cognitive impairment. *JAMA*. 2009;302(4):385–93, <http://dx.doi.org/10.1001/jama.2009.1064>.
- [92]. Visser PJ, Verhey F, Knol DL, Scheltens P, Wahlund L-O, Freund-Levi Y, et al. Prevalence and prognostic value of CSF markers of Alzheimer's disease pathology in patients with subjective cognitive impairment or mild cognitive impairment in the DESCRIPA study: a prospective cohort study. *Lancet Neurol*. 2009;8(7):619–27, [http://dx.doi.org/10.1016/s1474-4422\(09\)70139-5](http://dx.doi.org/10.1016/s1474-4422(09)70139-5).
- [93]. Vos SJB, Rossum IAvan, Verhey F, Knol DL, Soininen H, Wahlund L-O, et al. Prediction of Alzheimer disease in subjects with amnesic and nonamnesic MCI. *Neurology*. 2013;80(12):1124–32, <http://dx.doi.org/10.1212/wnl.0b013e318288690c>.
- [94]. Lemstra AW, de Beer MH, Teunissen CE, Schreuder C, Scheltens P, van der Flier WM, et al. Concomitant AD pathology affects clinical manifestation and survival in dementia with Lewy bodies. *J Neurol Neurosurg Psychiatry*. 2017;88(2):113, <http://dx.doi.org/10.1136/jnnp-2016-313775>.
- [95]. Johnson DK, Morris JC, Galvin JE. Verbal and visuospatial deficits in dementia with Lewy bodies. *Neurology*. 2005;65(8):1232–8, <http://dx.doi.org/10.1212/01.wnl.0000180964.60708.c2>.
- [96]. Tiraboschi P, Salmon DP, Hansen LA, Hofstetter RC, Thal LJ, Corey-Bloom J. What best differentiates Lewy body from Alzheimer's disease in early-stage dementia? *Brain*. 2006;129(3):729–35, <http://dx.doi.org/10.1093/brain/awh725>.
- [97]. Xu Y, Chen K, Zhao Q, Guo Q. Comparing the neuropsychological profiles of mild dementia with Lewy bodies and mild Alzheimer's disease. *Psychogeriatrics*. 2018;18(1):64–71, <http://dx.doi.org/10.1111/psyg.12293>.
- [98]. Guidi M, Paciaroni L, Paolini S, Padova SD, Scarpino O. Differences and similarities in the neuropsychological profile of dementia with Lewy bodies and Alzheimer's disease in the early stage. *J Neurol Sci*. 2006;248(1–2):120–3, <http://dx.doi.org/10.1016/j.jns.2006.05.017>.
- [99]. Peters F, Ergis A-M, Gauthier S, Dieudonné B, Verny M, Jolicoeur P, et al. Abnormal temporal dynamics of visual attention in Alzheimer's disease and in dementia with Lewy bodies. *Neurobiol Aging*. 2012;33(5):1012.e1–10, <http://dx.doi.org/10.1016/j.neurobiolaging.2011.10.019>.

- [100]. Brønne K, Breivik MH, Rongve A, Aarsland D. Neurocognitive deficits distinguishing mild dementia with Lewy bodies from mild Alzheimer's disease are associated with parkinsonism. *J Alzheimer's Dis.* 2016;53(4):1277–85, <http://dx.doi.org/10.3233/jad-160294>.
- [101]. Prentice JL, Schaeffer MJ, Wall AK, Callahan BL. A systematic review and comparison of neurocognitive features of late-life attention-deficit/hyperactivity disorder and dementia with Lewy bodies. *J Geriatr Psychol.* 2020;34(5):466–81, <http://dx.doi.org/10.1177/0891988720944251>.
- [102]. Mosimann UP, Mather G, Wesnes KA, O'Brien JT, Burn DJ, McKeith IG. Visual perception in Parkinson disease dementia and dementia with Lewy bodies. *Neurology.* 2004;63(11):2091–6, <http://dx.doi.org/10.1212/01.wnl.0000145764.70698.4e>.
- [103]. Bussè C, Anselmi P, Pompanin S, Zorzi G, Fragiaco F, Camporese G, et al. Specific verbal memory measures may distinguish Alzheimer's disease from dementia with Lewy bodies. *J Alzheimer's Dis.* 2017;1–7, <http://dx.doi.org/10.3233/jad-170154>. Preprint(Preprint).
- [104]. Petrova M, Pavlova R, Zhelev Y, Mehrabian S, Raycheva M, Traykov L. Investigation of neuropsychological characteristics of very mild and mild dementia with Lewy bodies. *J Clin Exp Neuropsychol.* 2016;38(3):354–60, <http://dx.doi.org/10.1080/13803395.2015.1117058>.
- [105]. John Joseph D, Nicholas MP, Mark D, Jose F, Eric G, Paul C. Intellectual, mnemonic, and frontal functions in dementia with lewy bodies: a comparison with early and advanced Parkinson's disease. *Behav Neurol.* 1999;11(3):173–83, <http://dx.doi.org/10.1155/1999/851860>.
- [106]. Smits LL, van Harten AC, Pijnenburg YAL, Koedam ELGE, Bouwman FH, Sistermans N, et al. Trajectories of cognitive decline in different types of dementia. *Psychol Med.* 2015;45(5):1051–9, <http://dx.doi.org/10.1017/s0033291714002153>.
- [107]. Ferman TJ, Smith GE, Boeve BF, Graff-Radford NR, Lucas JA, Knopman DS, et al. Neuropsychological differentiation of dementia with Lewy bodies from normal aging and Alzheimer's disease. *Clin Neuropsychologist.* 2006;20(4):623–36, <http://dx.doi.org/10.1080/13854040500376831>.
- [108]. Aldridge GM, Birnschein A, Denburg NL, Narayanan NS. Parkinson's disease dementia and dementia with Lewy bodies have similar neuropsychological profiles. *Front Neurol.* 2018;9:123, <http://dx.doi.org/10.3389/fneur.2018.00123>.
- [109]. Salmon DP, Heindel WC, Hamilton JM, Filoteo JV, Cidambi V, Hansen LA, et al. Recognition memory span in autopsy-confirmed dementia with Lewy bodies and Alzheimer's disease. *Neuropsychologia.* 2015;75:548–55, <http://dx.doi.org/10.1016/j.neuropsychologia.2015.07.014>.
- [110]. Bailon O, Roussel M, Boucart M, Krystkowiak P, Godefroy O. Psychomotor slowing in mild cognitive impairment, Alzheimer's disease and Lewy body dementia: mechanisms and diagnostic value. *Dement Geriatr Cogn.* 2010;29(5):388–96, <http://dx.doi.org/10.1159/000305095>.
- [111]. Perri R, Fadda L, Caltagirone C, Carlesimo GA. Word list and story recall elicit different patterns of memory deficit in patients with Alzheimer's disease, frontotemporal dementia, subcortical ischemic vascular disease, and lewy body dementia. *J Alzheimer's Dis.* 2013;37(1):99–107, <http://dx.doi.org/10.3233/jad-130347>.
- [112]. Ala T, Bakir D, Goel S, Feller N, Botchway A, Womack C. A mini-mental state examination formula may help to distinguish Alzheimer's disease from dementia with Lewy bodies. *J Alzheimer's Dis.* 2022;89(3):1119–29, <http://dx.doi.org/10.3233/jad-220392>.
- [113]. Cormack F, Aarsland D, Ballard C, Tovée MJ. Pentagon drawing and neuropsychological performance in dementia with Lewy Bodies, Alzheimer's disease, Parkinson's disease and Parkinson's disease with dementia. *Int J Geriatr Psychiatry.* 2004;19(4):371–7, <http://dx.doi.org/10.1002/gps.1094>.
- [114]. Ala TA, Hughes LF, Kyrouac GA, Ghobrial MW, Elble RJ. Pentagon copying is more impaired in dementia with Lewy bodies than in Alzheimer's disease. *J Neurology Neurosurg Psychiatry.* 2001;70(4):483, <http://dx.doi.org/10.1136/jnnp.70.4.483>.
- [115]. Harciarek M, Jodzio K. Neuropsychological differences between frontotemporal dementia and Alzheimer's disease: a review. *Neuropsychol Rev.* 2005;15(3):131–45, <http://dx.doi.org/10.1007/s11065-005-7093-4>.
- [116]. Quental NBM, Brucki SMD, Bueno OFA. Visuospatial function in early Alzheimer's disease: preliminary study. *Dement Neuropsychol.* 2009;3(3):234–40, <http://dx.doi.org/10.1590/s1980-57642009dn30300010>.
- [117]. Cronin-Golomb A, Corkin S, Rizzo JF, Cohen J, Growdon JH, Banks KS. Visual dysfunction in Alzheimer's disease: relation to normal aging. *Ann Neurol.* 1991;29(1):41–52, <http://dx.doi.org/10.1002/ana.410290110>.
- [118]. Monacelli AM, Cushman LA, Kavcic V, Duffy CJ. Spatial disorientation in Alzheimer's disease: the remembrance of things passed. *Neurology.* 2003;61(11):1491–7, <http://dx.doi.org/10.1212/wnl.61.11.1491>.
- [119]. Crowell TA, Luis CA, Cox DE, Mullan M. Neuropsychological comparison of Alzheimer's disease and dementia with Lewy bodies. *Dement Geriatr Cogn.* 2007;23(2):120–5, <http://dx.doi.org/10.1159/000097791>.
- [120]. Ota K, Murayama N, Kasanuki K, Kondo D, Fujishiro H, Arai H, et al. Visuo-perceptual assessments for differentiating dementia with Lewy bodies and Alzheimer's disease: illusory contours and other neuropsychological examinations. *Arch Clin Neuropsychol.* 2015;30(3):256–63, <http://dx.doi.org/10.1093/arclin/acv016>.
- [121]. Alegret M, Vinyes-Junqué G, Boada M, Martínez-Lage P, Cuberas G, Espinosa A, et al. Brain perfusion correlates of visuo-perceptual deficits in mild cognitive impairment and mild Alzheimer's disease. *J Alzheimer's Dis.* 2010;21(2):557–67, <http://dx.doi.org/10.3233/jad-2010-091069>.
- [122]. Alegret M, Boada-Rovira M, Vinyes-Junqué G, Valero S, Espinosa A, Hernández I, et al. Detection of visuo-perceptual deficits in preclinical and mild Alzheimer's disease. *J Clin Exp Neuropsychol.* 2009;31(7):860–7, <http://dx.doi.org/10.1080/13803390802595568>.
- [123]. Westervelt HJ, Bruce JM, Faust MA. Distinguishing Alzheimer's disease and dementia with Lewy bodies using cognitive and olfactory measures. *Neuropsychology.* 2016;30(3):304–11, <http://dx.doi.org/10.1037/neu0000230>.
- [124]. Monsch AU, Bondi MW, Butters N, Salmon DP, Katzman R, Thal LJ. Comparisons of verbal fluency tasks in the detection of dementia of the alzheimer type. *Arch Neurol.* 1992;49(12):1253–8, <http://dx.doi.org/10.1001/archneur.1992.00530360051017>.
- [125]. Henry JD, Crawford JR, Phillips LH. Verbal fluency performance in dementia of the Alzheimer's type: a meta-analysis. *Neuropsychologia.* 2004;42(9):1212–22, <http://dx.doi.org/10.1016/j.neuropsychologia.2004.02.001>.
- [126]. Henry JD, Crawford JR, Phillips LH. A meta-analytic review of verbal fluency deficits in Huntington's disease. *Neuropsychology.* 2005;19(2):243–52, <http://dx.doi.org/10.1037/0894-4105.19.2.243>.
- [127]. Rousseaux M, Sève A, Vallet M, Pasquier F, Mackowiak-Coudoliani MA. An analysis of communication in conversation in patients with dementia. *Neuropsychologia.* 2010;48(13):3884–90, <http://dx.doi.org/10.1016/j.neuropsychologia.2010.09.026>.
- [128]. Delbeuck X, Debachy B, Pasquier F, Moroni C. Action and noun fluency testing to distinguish between Alzheimer's disease and dementia with Lewy bodies. *J Clin Exp*

- Neuropsych. 2013;35(3):259–68, <http://dx.doi.org/10.1080/13803395.2013.763907>.
- [129]. Kramberger MG, Auestad B, Garcia-Plata S, Abdelnour C, Olmo JG, Walker Z, et al. Long-term cognitive decline in dementia with Lewy bodies in a large multicenter, international cohort. *J Alzheimer's Dis.* 2017;1–9, <http://dx.doi.org/10.3233/jad-161109>. Preprint(Preprint).
- [130]. Williams MM, Xiong C, Morris JC, Galvin JE. Survival and mortality differences between dementia with Lewy bodies vs Alzheimer disease. *Neurology.* 2006;67(11):1935–41, <http://dx.doi.org/10.1212/01.wnl.0000247041.63081.98>.
- [131]. Burn DJ, Rowan EN, Allan LM, Molloy S, O'Brien JT, McKeith IG. Motor subtype and cognitive decline in Parkinson's disease, Parkinson's disease with dementia, and dementia with Lewy bodies. *J Neurology Neurosurg Psychiatry.* 2006;77(5):585, <http://dx.doi.org/10.1136/jnnp.2005.081711>.
- [132]. Matar E, Martens KAE, Halliday GM, Lewis SJG. Clinical features of Lewy body dementia: insights into diagnosis and pathophysiology. *J Neurol.* 2020;267(2):380–9, <http://dx.doi.org/10.1007/s00415-019-09583-8>.
- [133]. Matar E, White SR, Taylor J-P, Thomas A, McKeith IG, Kane JPM, et al. Progression of clinical features in lewy body dementia can be detected over 6 months. *Neurology.* 2021;97(10):e1031–40, <http://dx.doi.org/10.1212/wnl.00000000000012450>.
- [134]. Biundo R, Weis L, Bostantjopoulou S, Stefanova E, Falup-Pecurariu C, Kramberger MG, et al. MMSE and MoCA in Parkinson's disease and dementia with Lewy bodies: a multicenter 1-year follow-up study. *J Neural Transmission.* 2016;123(4):431–8, <http://dx.doi.org/10.1007/s00702-016-1517-6>.
- [135]. Economou A, Routsis C, Papageorgiou SG. Episodic memory in Alzheimer disease, frontotemporal dementia, and dementia with Lewy bodies/Parkinson disease dementia: disentangling retrieval from consolidation. *Alzheimer Dis Assoc Disord.* 2015;30(1):47–52, <http://dx.doi.org/10.1097/wad.0000000000000089>.
- [136]. Bussè C, Caffarra P, Rossi A, Zorzi G, Fragiaco F, Camporese G, et al. Testing hippocampal memory in prodromal dementia with Lewy bodies. *J Alzheimer's Dis.* 2018;1–5, <http://dx.doi.org/10.3233/jad-180166>. Preprint(Preprint).
- [137]. Patel N, Peterson KA, Ralph ML, Rowe JB, Patterson K, Cappa S, et al. The Mini Linguistic State Examination (MLSE): a standardised tool to classify and monitor primary progressive aphasia. *Alzheimer's Dementia.* 2020;16, <http://dx.doi.org/10.1002/alz.040853>.
- [138]. Matias-Guiu JA, Pytel V, Hernández-Lorenzo L, Patel N, Peterson KA, Matias-Guiu J, et al. Spanish version of the mini-linguistic state examination for the diagnosis of primary progressive aphasia. *J Alzheimer's Dis Jad.* 2021;83(2):771–8, <http://dx.doi.org/10.3233/jad-210668>.
- [139]. Yamada Y, Shinkawa K, Nemoto M, Ota M, Nemoto K, Arai T. Speech and language characteristics differentiate Alzheimer's disease and dementia with Lewy bodies. *Alzheimer's Dementia Diagnosis Assess Dis Monit.* 2022;14(1):e12364, <http://dx.doi.org/10.1002/dad2.12364>.
- [140]. Burton EJ, Karas G, Paling SM, Barber R, Williams ED, Ballard CG, et al. Patterns of cerebral atrophy in dementia with Lewy bodies using voxel-based morphometry. *Neuroimage.* 2002;17(2):618–30, <http://dx.doi.org/10.1006/nimg.2002.1197>.
- [141]. Karas GB, Burton EJ, Rombouts SARB, van Schijndel RA, O'Brien JT, Scheltens PH, et al. A comprehensive study of gray matter loss in patients with Alzheimer's disease using optimized voxel-based morphometry. *Neuroimage.* 2003;18(4):895–907, [http://dx.doi.org/10.1016/s1053-8119\(03\)00041-7](http://dx.doi.org/10.1016/s1053-8119(03)00041-7).
- [142]. Burton EJ, Barber R, Mukaetova-Ladinska EB, Robson J, Perry RH, Jaros E, et al. Medial temporal lobe atrophy on MRI differentiates Alzheimer's disease from dementia with Lewy bodies and vascular cognitive impairment: a prospective study with pathological verification of diagnosis. *Brain.* 2009;132(1):195–203, <http://dx.doi.org/10.1093/brain/awn298>.
- [143]. Lebedev AV, Westman E, Beyer MK, Kramberger MG, Aguilar C, Pirtosek Z, et al. Multivariate classification of patients with Alzheimer's and dementia with Lewy bodies using high-dimensional cortical thickness measurements: an MRI surface-based morphometric study. *J Neurol.* 2013;260(4):1104–15, <http://dx.doi.org/10.1007/s00415-012-6768-z>.
- [144]. Ballmaier M, O'Brien JT, Burton EJ, Thompson PM, Rex DE, Narr KL, et al. Comparing gray matter loss profiles between dementia with Lewy bodies and Alzheimer's disease using cortical pattern matching: diagnosis and gender effects. *Neuroimage.* 2004;23(1):325–35, <http://dx.doi.org/10.1016/j.neuroimage.2004.04.026>.
- [145]. Barber R, Ballard C, McKeith IG, Gholkar A, O'Brien JT. MRI volumetric study of dementia with Lewy bodies: a comparison with AD and vascular dementia. *Neurology.* 2000;54(6):1304–9, <http://dx.doi.org/10.1212/wnl.54.6.1304>.
- [146]. Galvin J, Price J, Yan Z, Morris J, Sheline Y. Resting bold fMRI differentiates dementia with Lewy bodies vs. Alzheimer's disease. *Alzheimer's Dementia.* 2011;7(4):e69, <http://dx.doi.org/10.1016/j.jalz.2011.09.147>.
- [147]. Kenny ER, O'Brien JT, Firbank MJ, Blamire AM. Subcortical connectivity in dementia with Lewy bodies and Alzheimer's disease. *Br J Psychiat.* 2013;203(3):209–14, <http://dx.doi.org/10.1192/bjp.bp.112.108464>.
- [148]. Gomperts SN, Locascio JJ, Makaretz SJ, Schultz A, Caso C, Vasdev N, et al. Tau positron emission tomographic imaging in the Lewy body diseases. *JAMA Neurol.* 2016;73(11):1334, <http://dx.doi.org/10.1001/jamaneurol.2016.3338>.
- [149]. Brooks DJ. Imaging amyloid in Parkinson's disease dementia and dementia with Lewy bodies with positron emission tomography. *Mov Disord.* 2009;24(S2):S742–7, <http://dx.doi.org/10.1002/mds.22581>.
- [150]. Bohnen NI, Müller MLTM, Frey KA. Molecular imaging and updated diagnostic criteria in Lewy body dementias. *Curr Neurol Neurosci.* 2017;17(10):73, <http://dx.doi.org/10.1007/s11910-017-0789-z>.
- [151]. Marquie M, Verwer EE, Meltzer AC, Kim SJW, Agüero C, Gonzalez J, et al. Lessons learned about [F-18]-AV-1451 off-target binding from an autopsy-confirmed Parkinson's case. *Acta Neuropathologica Commun.* 2017;5(1):75, <http://dx.doi.org/10.1186/s40478-017-0482-0>.
- [152]. Metzler-Baddeley C, Baddeley RJ, Lovell PG, Laffan A, Jones RW. Visual impairments in dementia with Lewy bodies and posterior cortical atrophy. *Neuropsychology.* 2010;24(1):35–48, <http://dx.doi.org/10.1037/a0016834>.
- [153]. Nedelska Z, Josephs KA, Graff-Radford J, Przybelski SA, Lesnick TG, Boeve BF, et al. 18F-AV-1451 uptake differs between dementia with Lewy bodies and posterior cortical atrophy. *Mov Disord.* 2019;34(3):344–52, <http://dx.doi.org/10.1002/mds.27603>.
- [154]. Boes S, Botha H, Machulda M, Lowe V, Graff-Radford J, Whitwell JL, et al. Dementia with Lewy bodies presenting as Logopenic variant primary progressive Aphasia. *Neurocase.* 2020;26(5):259–63, <http://dx.doi.org/10.1080/13554794.2020.1795204>.
- [155]. Teichmann M, Migliaccio R, Kas A, Dubois B. Logopenic progressive aphasia beyond Alzheimer's—an evolution towards dementia with Lewy bodies. *J Neurol Neu-*

- rosurg Psychiatry. 2013;84(1):113, <http://dx.doi.org/10.1136/jnnp-2012-302638>.
- [156]. Youmans G, Bourgeois M. Theory of mind in individuals with Alzheimer-type dementia. *Aphasiology*. 2010;24(4):515–34, <http://dx.doi.org/10.1080/02687030902801573>.
- [157]. Lucena ATD, Bhalla RK, Santos TTBAD, Dourado MCN. The relationship between theory of mind and cognition in Alzheimer's disease: a systematic review. *J Clin Exp Neuropsych*. 2020;42(3):223–39, <http://dx.doi.org/10.1080/13803395.2019.1710112>.